Lineage from Two Snapshots, Learned without Lineage Labels Supplementary Material

Anonymous CVPR submission

Paper ID *****

A. Implementation details

All values are those of our code. OIAM frames are one minute apart; elsewhere H is converted to minutes before entering any model.

LineageRAFT tokens. From a cell's mask, the eigendecomposition of its pixel covariance gives the unit major axis $\mathbf{u}$ and standard deviations $\sigma_{\parallel} \geq \sigma_{\perp}$. Let s be the median nearest-neighbour distance among targets, N_a, N_b the counts and $\boldsymbol{\mu}_a, \boldsymbol{\mu}_b$ the centroids. Targets have canonical positions $\tilde{\mathbf{y}}_j = (\mathbf{y}_j - \boldsymbol{\mu}_b)/s$; ancestors $\tilde{\mathbf{x}}_i = \sqrt{N_b/N_a}(\mathbf{x}_i - \boldsymbol{\mu}_a)/s$ (radial) or $(\mathcal{A}(\mathbf{x}_i) - \boldsymbol{\mu}_b)/s$ with $\mathcal{A}$ the Darcy advection; this is also $\mathbf{c}_i^{(0)}$. The 19-d token concatenates the canonical position (2), $(\mathbf{x}-\text{image centre})/10s$ (2), $(u_y^2-u_x^2, 2u_yu_x)$ (2), $\log\sigma_{\parallel}-2$ and $\log\sigma_{\perp}-1.4$ (2), $\tanh(\mathbf{b}/4)$ (2) and $\log(1+\|\mathbf{b}\|)$ (1) for $\mathbf{b}$ the shortest axis-aligned vector to the image border in units of s, the drift $(\boldsymbol{\mu}_b-\boldsymbol{\mu}_a)/10s$ (2), $[\log_2 \frac{H}{60}, \log\frac{N_b}{N_a}, \log N_a, \log N_b]$ (4) and a one-hot ancestor/target flag (2). Shape-free models zero the four shape entries. Augmentation applies one matrix to all vectors: *full*, a rotation $\theta \sim U[0, 2\pi)$ then a reflection with probability $\frac{1}{2}$; *flip*, reflecting x with probability $\frac{1}{2}$; *none*.

LineageRAFT layers (d=128, 2,875,659 parameters). Input MLPs (one per snapshot): Linear$(19, d)$–GELU–Linear(d, d). *LocalAttn* is kNN attention with relative-position encoding: queries $W_q\text{LN}(\mathbf{h}^q)$; keys and values $W_{kv}\text{LN}(\mathbf{h}^x)$ gathered at the neighbours plus $\phi(\boldsymbol{\delta}/4)$, with ϕ = Linear$(2, d)$–GELU–Linear$(d, 2d)$ of the neighbour offset $\boldsymbol{\delta}$; 4 heads of 32 channels, softmax over neighbours; residual output projection, then a residual FFN Linear$(d, 2d)$–GELU–Linear$(2d, d)$ without normalisation. *Encoder*: two LocalAttn layers per snapshot over the k_{self}+1=13 nearest cells (self included) in canonical coordinates. *GlobalCtx*: $\mathbf{h} \leftarrow \mathbf{h} + \text{MLP}([\mathbf{h}, \overline{\mathbf{h}}])$, $\overline{\mathbf{h}}$ the snapshot mean, MLP Linear$(2d, d)$–GELU–Linear(d, d). *Iteration* t=1..4 (unshared weights), with $\mathbf{c} = \text{sg}(\mathbf{c}^{(t-1)})$ (stop-gradient): (i) ancestor i attends to the k_{cross}=24 targets nearest $\mathbf{c}_i$, offsets $\tilde{\mathbf{y}}_j - \mathbf{c}_i$; (ii) ancestors attend over their fixed 13-NN graph of initial canonical positions; (iii) $\mathbf{c}_i^{(t)} = \mathbf{c}_i + \Delta_t(\mathbf{h}_i)$, Δ_t Linear$(d, 2)$ with zero-initialised weight and bias; (iv) target j attends to its 24 nearest $\text{sg}(\mathbf{c}^{(t)})$, offsets $\mathbf{c}_i - \tilde{\mathbf{y}}_j$. *Assignment*: target j is scored against its k_{out}=24 nearest final centres by Linear$(2d+3, d)$–GELU–Linear(d, d)–GELU–Linear$(d, 1)$ on $[\mathbf{h}_j, \mathbf{h}_i, \mathbf{o}/4, \|\mathbf{o}\|/4]$, $\mathbf{o} = \tilde{\mathbf{y}}_j - \mathbf{c}_i^{(T)}$ (not detached), plus a dustbin logit Linear$(d, 1)(\mathbf{h}_j)$; softmax over 25. At test time the probabilities are scattered into an $N_b \times N_a$ matrix, the dustbin dropped and rows renormalised; no test-time augmentation. *Family head*: Linear$(2d+3, d)$–GELU–Linear$(d, 1)$ on $[\mathbf{h}_j, \mathbf{h}_k, \boldsymbol{\delta}/4, \|\boldsymbol{\delta}\|/4]$, $\boldsymbol{\delta} = \tilde{\mathbf{y}}_k - \tilde{\mathbf{y}}_j$, logits averaged over both orders.

Labels and losses. A target's label is the track alive at a reached through its parents; it is the dustbin if the chain ends at a root born after a or that track has no detection at a (in RODSIM, e.g. outside the field of view). Cross-entropy runs over the 25 classes and ignores targets whose labelled ancestor is not among their candidates. With $\bar{\mathbf{g}}_i$ the mean canonical position of the labelled descendants of ancestor i, $\mathcal{L}_c = \sum_{t=1}^{T} 0.8^{T-t} \,\text{mean}_i \min(\|\mathbf{c}_i^{(t)} - \bar{\mathbf{g}}_i\|, 10)$. The family loss is a binary cross-entropy for "same ancestor" on the directed 8-NN graph of $\tilde{\mathbf{y}}$, restricted to pairs of labelled targets (skipped if fewer than three are labelled). With P the dense $N_b \times (N_a+1)$ assignment matrix, dustbin included, the consistency loss on the same pairs is $\text{BCE}(\text{clip}(\sum_i P_{ji}P_{ki}, 10^{-5}, 1-10^{-5}), [\text{same}])$. In total, $\mathcal{L} = \mathcal{L}_{\text{CE}} + 0.3\mathcal{L}_c + 0.5\mathcal{L}_{\text{fam}} + \lambda_{\text{cons}}\mathcal{L}_{\text{cons}}$.

Training. A batch is one window: a movie or colony drawn uniformly, $H = \text{round}(\exp U[\ln h_{\min}, \ln h_{\max}])$, and a uniform frame a with at least two cells, redrawn until frame $a+H$ exists; all cells of both frames are used. We use AdamW (weight decay 0.01), OneCycle with 5% warm-up and otherwise PyTorch defaults (cosine, start and end at

	OIAM main	RODSIM	fine-tune
data	M0–M2	256 colonies	pseudo-lineage
initial weights	random	random	RODSIM
steps	80k	30k	6k
peak learning rate	$5{\cdot}10^{-4}$	$5{\cdot}10^{-4}$	$2{\cdot}10^{-4}$
$h_{\min}$–$h_{\max}$ (min)	8–240	8–240	8 or 15–240
$\mathbf{c}^{(0)}$	Darcy	radial	radial
augmentation	none	full	full
shape features	yes	no	no
λ_{cons}	0.5	0	0
seeds	0, 1, 2	0, 1, 2	0

Table S1. **LineageRAFT settings.** All use T=4, k_{self}=12, k_{cross}=k_{out}=24. The annotation-free OIAM model uses the main settings on pseudo-lineage; the ablation without consistency sets λ_{cons}=0. Fine-tuning: on OIAM $h_{\min}$=8, on external sources 15.

$1/25$ and $1/(25{\cdot}10^4)$ of the peak, $\beta_1 \in [0.85, 0.95]$), gradient clipping at norm 1 and bf16 autocast (Table S1).

OIAM evaluation windows. With C=66 frames the median duration of tracks that begin and end with a division in M0–M2, horizons are $\{C/2, C, 2C\}$ and anchor frames lie on a grid of step $C/2$=33 from the first frame with at least 8 cells, keeping windows with $a{+}H$ within the movie: 20/19/17 windows on M3 and 19/18/16 on M4 at 33/66/132 minutes (109). Only frames a and b are given (annotated centres in a seeded random order, and mask moments). The field of interest is the whole image. A target is scored if its chain reaches a track alive and annotated at a; descendants of roots born after a (entries), broken chains and unannotated ancestors are excluded. Accuracy is averaged over windows; *dense* windows have more than 200 ancestors.

Darcy prior. With s_a the median nearest-neighbour spacing of ancestors, the grid step is $\ell = \max(s_a/2, 2\,\text{px})$ over the image. For each snapshot, grid cells containing a centre are dilated $r = \max(1, \text{round}(1.5 s_a/\ell))$ times, hole-filled and eroded $r{-}1$ times. We solve $4p_m - \sum_{n\sim m} p_n = g\ell^2$ with p=0 outside the domain and grid (sparse direct solver); $\mathbf{v} = -\nabla p$ by central differences, interpolated bilinearly. By default $g = \ln 2/68\,\text{min}^{-1}$ (a fixed doubling time, not fitted). If both snapshots are fully visible (at most 2% of cells within $2s_a$ of the border), $g = \ln(\max(N_b/N_a, 1))/H$ and the result is re-centred on $\boldsymbol{\mu}_b$; LineageRAFT uses this rule, while the "Darcy 68" baseline always uses the fixed rate without re-centring and assigns each target to the nearest advected ancestor. Advection: 12 midpoint (RK2) steps of $H/12$ in $(1{-}\tau)\mathbf{v}_a{+}\tau\mathbf{v}_b$, τ the elapsed fraction.

Genealogy network (1,291,910 parameters). Pairs are the undirected union of each cell's 8 nearest neighbours. With D the pair distance, ς a cell's nearest-neighbour distance and $\hat{\mathbf{d}}$ the pair direction, the ten standardised features are $D/\sqrt{\varsigma_j\varsigma_k}$, $D/\sqrt{\sigma_{\|j}\sigma_{\|k}}$, $|\mathbf{u}_j{\cdot}\mathbf{u}_k|$, $|\hat{\mathbf{d}}{\cdot}\mathbf{u}_j|$, $|\hat{\mathbf{d}}{\cdot}\mathbf{u}_k|$, $|\log\frac{\sigma_{\|j}}{\sigma_{\|k}}|$, $|\log\frac{\sigma_{\perp j}}{\sigma_{\perp k}}|$, the difference of log aspect ratios, $|\log\frac{\varsigma_j}{\varsigma_k}|$ and the difference of distances to the centroid over their RMS (all absolute). The z-scored frame is sampled bilinearly on a 64×64 grid at 1 px, x axis along the pair, centred on its midpoint; a second channel holds $\exp(-\|\mathbf{p}-\mathbf{p}_c\|^2/0.01)$ at both centres (patch coordinates in $[-1, 1]$). CNN: four stages of [conv3×3–BN–ReLU]×2 (32/64/128/256 channels), 2×2 max-pooling between stages, global average pooling; geometry MLP 10→128→128; head 384→256→6 (ReLU). The outputs are trained with mean binary cross-entropy on $\tau_{jk} < \{8, 16, 33, 66, 132, 264\}$, τ_{jk} being b minus the birth frame of the daughter on j's side of the last common ancestor (10^4 if none). Data: M0–M2 frames $250, 252, \ldots, 798$ with at least 8 cells; batches of 4 frames × 256 pairs; pair order reversed (per frame) and the perpendicular axis reflected (per pair) with probability $\frac{1}{2}$; AdamW (lr 10^{-3}, weight decay 10^{-4}), OneCycle (PyTorch defaults), 15k steps, bf16. At test time the logits of both orders are averaged and w_{jk} is the sigmoid of the output whose threshold equals H; thresholds 8, 16 and 264 are auxiliary targets only. The annotation-free variant sets $\tau{=}H{-}0.5$ for the smallest threshold H at which both cells chain back to the same pseudo-ancestor. *Potts decoding*: $q \leftarrow p$ (dustbin removed, renormalised, floored at 10^{-9}), then 10 synchronous updates $q_j \leftarrow \text{softmax}(\log p_j + \beta\sum_{k\in\mathcal{N}(j)}(2w_{jk}{-}1)q_k)$ on the same graph, without degree normalisation; β=2 (main) or 1 (label-free table). Ensembles average probabilities before decoding. Windows with fewer than 9 targets or 2 ancestors are not decoded.

RODSIM. Cells are spherocylinders (cap radius r=0.5 μm, length L, angle θ), each represented by $\max(2, \lceil(L{-}2r)/r\rceil{+}1)$ spheres along its axis. Every minute: (1) $L \leftarrow Le^{\lambda}$, $\lambda = \frac{\ln 2}{T_d}e^{\epsilon}$, $\epsilon \sim \mathcal{N}(0, \text{CV}_g^2)$ drawn at birth; (2) adder division when $L{-}L_{\text{birth}} \geq \ell_0 e^{\eta}$, $\eta \sim \mathcal{N}(0, \text{CV}_a^2)$ drawn at birth: the septum sits at fraction $f = \text{clip}(\frac{1}{2}{+}\mathcal{N}(0, \sigma_f^2), 0.3, 0.7)$ and daughters of lengths $fL, (1{-}f)L$ lie on either side along $\theta \mp \varsigma\alpha/2 + \mathcal{N}(0, \sigma_\theta^2)$, $\varsigma{=}\pm 1$, $\alpha = |\mathcal{N}(0, (1.2533\bar{\alpha})^2)|$ (mean $\bar{\alpha}$); widths inherit a factor $e^{\mathcal{N}(0, \text{CV}_w^2)}$; (3) eight sub-steps in which spheres of different cells closer than $2r$ repel with force $2r{-}d$, giving per cell $\Delta\mathbf{x} = (\kappa\mathbf{F}_{\|}{+}\mathbf{F}_{\perp})/2L$ and $\Delta\theta = M/(L^3/12)$ for net force $\mathbf{F}$ and torque M; (4) $\Delta\mathbf{x} \sim \mathcal{N}(0, D^2 I)$, $\Delta\theta \sim \mathcal{N}(0, D_r^2)$. Founders: positions $\mathcal{N}(0, (2\,\mu\text{m})^2 I)$, $\theta \sim U[0, \pi)$, $L = \ell_0 U(1, 1.9)$, $L_{\text{birth}} = L/U(1, 1.9)$. Colonies stop at $N_{\max}$ cells or 1,500 minutes; every minute is recorded with $\sigma_{\|} = L/\sqrt{12}$, $\sigma_{\perp} = wr/2$. *Field of view*: with R = 1.4× the final RMS radius +1 μm, a square of half-side ρR centred at the final centroid $+\,\mathbf{o}R$, at γ px/μm; cells outside are dropped, ancestry uses the full

parameter	range (uniform)	midpoint
birth length ℓ_0	1–3 μm	2
doubling time T_d	40–110 min	75
growth-rate CV$_g$	0.05–0.3	0.15
adder CV$_a$	0.05–0.25	0.15
division angle noise σ_θ	0–0.15 rad	0.07
position noise D	0–0.02 μm/min	0.01
angle noise D_r	0–0.03 rad/min	0.015
drag anisotropy κ	0.5–2	1
founders n_0	1–4	–
final cells $N_{\max}$	900–2199	–
pixel scale γ	6–14 px/μm	–
FOV half-size ρ	0.35–1.2	–
FOV offset $\mathbf{o}$ (per axis)	−0.4–0.4	–
division modes only (otherwise 0)		
septum s.d. σ_f	0–0.08	–
mean V-snap angle $\bar{\alpha}$	0–1.2 rad	–
width CV$_w$	0–0.06	–
length/width noise σ_{obs}	0–0.08	–
axis noise σ_{ang}	0–0.15 rad	–

Table S2. **RODSIM randomisation**, drawn once per colony (integers for $n_0, N_{\max}$). Midpoints: fixed-physics ablation.

tree. Division modes add per-frame noise $e^{\mathcal{N}(0,\sigma^2_{\text{obs}})}$ on $\sigma_\parallel, \sigma_\perp$ and $\mathcal{N}(0, \sigma^2_{\text{ang}})$ on the axis (Table S2). Colony seeds are 0..255. *Rendering* (RODSIM genealogy; 6 frames per colony at minute ≥200 with 30–1,500 cells, 200 colonies): rescale so that the width $4\,\text{median}\,\sigma_\perp$ is $U(12, 19)$ px; draw capsules of radius $2\sigma_\perp$ through their signed distance, blending background $U(0.6, 1)$ and cell level $U(0, 0.3)$ by $\text{sigmoid}(-\text{sdf}/U(0.5, 1.5))$, with per-cell factor $1{+}\mathcal{N}(0, U(0, 0.15)^2)$; add an outside halo (amplitude $U(0, 0.6)$, decay $U(1, 4)$ px), a linear gradient $\mathcal{N}(0, 0.15^2)$ per axis, Gaussian blur $U(0.7, 2)$ px and noise of s.d. $U(0.01, 0.08)$. Render v2 instead inherits the per-cell factor from the mother times $e^{\mathcal{N}(0,U(0.03,0.12)^2)}$ and constricts the longest $U(5, 25)\%$ of cells at mid-cell (depth $U(0.15, 0.45)r$, width $0.6r$). Image patches are also z-scored per patch.

Self-labelling. For frames $t \to t{+}1$, each cell of t offers a *stay* slot at $\mathbf{x}_i$ and two daughter slots at $\mathbf{x}_i \pm \frac{L_i}{4}\mathbf{u}_i$ ($L_i = \sqrt{12}\sigma_{\parallel i}$), and each cell of $t{+}1$ a private no-parent column. Costs are Euclidean distances, plus 10 px for daughter slots; no parent costs 60 px; only the 8 nearest slots of each type are finite. Each cell of $t{+}1$ is matched to one column by exact linear assignment (SciPy); a slot takes one cell, so a parent can take up to three. Without masks, daughter slots sit at $\mathbf{x}_i$, the extra cost is $0.55s$ and no parent $3.5s$ (s the median nearest-neighbour spacing of frame t). Pseudo-ancestry composes links backwards; a no-parent link gives the dustbin. *Temporal bootstrapping* replaces each link by the argmax (dustbin excluded) of the RODSIM model on $(t, t{+}1)$, positions only, with H the frame interval. For adaptation, each source's sequences, sorted by name, alternate between two folds; a model fine-tuned on one fold is tested on the other.

Per-colony processing and SV. Ancestors are grouped by single linkage at 3 median spacings; targets join the colony of their nearest ancestor; the RODSIM model runs per colony (a one-ancestor colony takes all its targets). SV scores nearest assignment and per-colony LineageRAFT by *coverage*, the fraction of ancestors receiving at least one target (all targets counted), and keeps the higher; ties go to the lower coefficient of variation of family sizes (zeros included), then to nearest assignment. External inputs are positions only, with image size $1.05\times$ the detections' bounding box $+5$ px.

Compute. Training ran on one shared NVIDIA H200: 1.9–3.0 h per 80k-step OIAM model, 0.6–1.2 h per 30k-step RODSIM model, about 13 min per 6k-step fine-tune; 19 min for the OIAM genealogy network (15k steps) and about 41 min for the RODSIM one, including rendering.

B. What family registration needs

Tab. S3 isolates the family centre. At a fixed budget of 30k steps, accuracy rises with the number of refinement rounds (0.79, 0.83, 0.86 and 0.88 at 33 minutes for $T{=}0, 1, 2, 4$), and keeping all four attention rounds but freezing the centre (the direct matcher) falls back to the level of the matcher without cross-attention, most clearly on dense windows (0.63 *vs*. 0.77 at 33 minutes, 0.36 *vs*. 0.47 at 66). The gain is largest at short and medium gaps and small at 132 minutes (0.419 *vs*. 0.424), where the partition limits accuracy. The training logs show that the true ancestor is among a target's 24 candidates for 99% of training targets with a moving centre and for 95% with a fixed one; the controls below show that this is not the explanation. Longer training matters as much as the architecture: the same model (seed 0) reaches 0.877 at 33 minutes after 30k steps and 0.962 after 80k. Longer training does not close the gap. In a second protocol, fixed before training, we trained the direct matcher at the main-table budget (80k steps, three seeds) with Darcy and with radial initialisation, and compared seed-mean accuracy per window. With Darcy initialisation, LineageRAFT is ahead by 0.101 [0.068, 0.138], 0.096 [0.072, 0.123] and 0.017 [0.004, 0.031] at 33/66/132 minutes, and on dense windows by 0.201, 0.169 and 0.061, winning every dense window at every gap (hypotheses D1, D2 hold). With radial initialisation it is ahead by 0.102 and 0.068 at 33 and 66 minutes but ties at 132 (−0.005 [−0.027, 0.019]; D3 fails there), although it still leads on dense windows (0.069 [0.030, 0.108]). Candidate recall in training stays at 0.95–0.96 for the direct matcher and 0.997 for LineageRAFT af-

	All windows			Dense (>200)		
Model	33	66	132	33	66	132
30k steps, seed 0 (core ablation)						
T=4, moving centre (reference)	.877	.663	.424	.773	.469	.208
T=2	.858	.664	.431	.732	.467	.217
T=1	.825	.648	.422	.681	.435	.206
T=0: learned pairwise matcher	.790	.620	.416	.653	.405	.196
T=4, fixed centre (direct matcher)	.806	.623	.419	.631	.364	.191
80k steps, mean of three seeds (matched direct matcher)						
LineageRAFT, Darcy init.	.957	.780	.473	.907	.598	.267
Direct matcher, Darcy init.	.856	.684	.456	.706	.429	.207
LineageRAFT, radial init.	.948	.748	.454	.894	.567	.275
Direct matcher, radial init.	.846	.681	.458	.685	.433	.206
Direct matcher, 96 candidates	.868	.696	.458	.736	.475	.229
Direct matcher + centre loss	.904	.709	.457	.789	.445	.205
LineageRAFT, T=1	.873	.670	.450	.756	.466	.233
80k steps, seed 0						
LineageRAFT without consistency term	.963	.789	.478	.918	.618	.269
LineageRAFT, T=8	.941	.750	.441	.885	.582	.237

Table S3. **What family registration needs** (OIAM M3/M4, accuracy, argmax). All rows use shape features, no rotation augmentation and the consistency term; Darcy initialisation unless stated. T: rounds of cross-snapshot attention and centre update. With T=0 each snapshot is encoded on its own and targets are scored against the initial anchor positions, a learned pairwise matcher. The direct matcher is the same network with the same 2.9M parameters, data, losses and schedule, except that the family centre never moves and has no loss: targets cross-attend to, and are scored against, candidate ancestors at their initial positions. Controls: the direct matcher with 96 instead of 24 candidate ancestors per target, or with the centre head trained by the centre loss but never used. All three protocols were fixed before training and run once.

ter 80k steps. A third protocol, also fixed before training (three seeds each, Darcy initialisation, 80k steps), tested three simpler explanations of the gap. (K1) Four times as many candidates (96) restore the direct matcher's recall to 0.99 but its accuracy only to 0.868/0.696/0.458; LineageRAFT stays ahead by 0.089 [0.062, 0.117] and 0.084 [0.067, 0.102] at 33 and 66 minutes, so missing candidates are not the cause. (K2) Training the centre head with the centre loss but never using it gives 0.904/0.709/0.457: centre supervision as an auxiliary task explains about half of the gain at 33 minutes and little at 66 (LineageRAFT ahead by 0.053 [0.026, 0.085] and 0.071 [0.044, 0.100]). (K3) One refinement round instead of four, with the same full recall, gives 0.873/0.670/0.450 (LineageRAFT ahead by 0.083 [0.062, 0.107] at 33 minutes). All three hypotheses hold, and LineageRAFT wins every dense window against every control at every gap; at 132 minutes the differences are 0.015–0.023. What matters is that candidates are scored relative to a centre that has moved over several rounds, not which ancestors are in the candidate set.

How do the centres move? Fig. S1 follows the family centres of the full model (seed 0) through the four rounds on M3/M4. The first round removes most of the initial error and later rounds keep reducing it: on dense windows the mean distance to the true family centroid falls from 2.1 to 0.3 cell spacings at 33 minutes, from 3.7 to 1.3 at 66 minutes and from 6.5 to 3.1 at 132 minutes, and the true ancestor is among a target's 24 nearest centres for 0.99/0.97/0.90 of targets instead of 0.90/0.78/0.69 at initialisation. Candidate recall alone does not explain the gain: the direct matcher with 96 candidates reaches 0.99 recall in training and still trails by 0.08–0.09 at 33 and 66 minutes (Tab. S3). What helps is that target offsets are measured from a point close to the family. Accuracy follows the centre: targets whose family centre ends within half a spacing of the true centroid are assigned correctly in 0.98/0.89/0.62 of cases at 33/66/132 minutes, beyond four spacings in 0.05–0.07. Larger families are harder: at 132 minutes accuracy is 0.35, 0.33 and 0.25 for families of 2, 3–4 and 5–8 scored descendants (rare unsplit cells: 0.12).

On small colonies the centre adds little. Under a protocol fixed before training (`docs/vv_family_vs_direct_protocol.md`) we repeated the bootstrapped self-labelling of blind test 4 (main-text Table 5) with the direct matcher: initialised from the RODSIM direct matcher, trained on the same pseudo-lineage of the other fold with the same settings, and evaluated on the same windows of the 28 van Vliet colonies. Self-labelling helps both equally (0.660→0.698 for LineageRAFT, 0.655→0.695 for the direct matcher), and LineageRAFT leads by only 0.003 [−0.001, 0.007] over colonies (17 of 28 colonies; ahead in all three subsets by 0.002–0.005): the primary hypothesis, a CI above zero, fails. These windows are small (median 10 ancestors, 99th percentile 186, against more than 200 for a dense OIAM window); post hoc, in the 188 windows with more than 100 ancestors at 30 minutes LineageRAFT leads by 0.028 [0.023, 0.033] (window bootstrap), and by at most 0.004 elsewhere. The same holds on OIAM (post hoc, `experiments/centre_crowding_analysis.py`): the full model's margin over the direct matcher grows with the number of ancestors in a window (Spearman ρ 0.88/0.88/0.69 at 33/66/132 minutes), from 0.02/0.02/−0.01 in windows of at most 25 ancestors to 0.24/0.19/0.08 in windows of more than 400. The moving family centre is thus a remedy for crowding: it matters where many families compete for each target, as in dense OIAM windows, and not in microcolonies, where a learned matcher does as well.

C. All in-domain variants

Tab. S4 extends Table 2 of the main paper with Trackastra's greedy decoding, the rotation-augmented model with

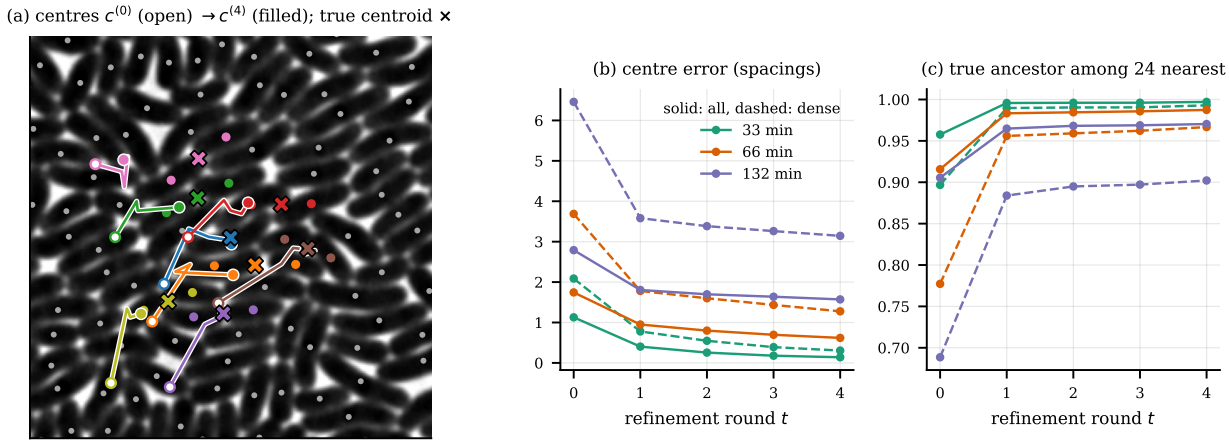


Figure S1. **Family centres through the refinement rounds.** (a) The eight families nearest the centre of a crop of a dense 66-minute OIAM window (the window of the qualitative figure) among those whose initial (Darcy) centre is more than 1.5 spacings off; dots: their scored descendants. (b) Mean distance from each ancestor's centre to its true family centroid, in target cell spacings, after round t (t=0: initialisation). (c) Fraction of targets whose true ancestor is among the 24 centres nearest to them, the candidate set of the assignment step. Solid: all M3/M4 windows; dashed: dense windows (N_a>200).

Darcy initialisation, the full model with the genealogy decoder, and the undecoded three-seed ensemble.

D. Leave-one-movie-out evaluation

OIAM M3/M4 were inspected during development. Tab. S5 trains on four OIAM movies and tests on the fifth, for all five folds. The full model, retrained in each fold under a protocol fixed beforehand (`docs/full_lomo_protocol.md`), reaches 0.954/0.792/0.483 on average over held-out movies, against 0.957/0.780/0.473 on the M3/M4 test: the development test movies did not flatter it. All three hypotheses hold: it beats the 30k-step no-rotation model at 33 and 66 minutes and the regressor at every gap on all five folds, and averages at least 0.90 at 33 minutes. On dense windows it averages 0.90/0.60/0.26.

Does the no-rotation gain need the chamber orientation? In-domain OIAM models are trained without rotation augmentation, which exploits the fixed orientation of the growth chambers in the image. We tested whether this prior must be known (`docs/orientation_protocol.md`): each M3/M4 window is presented in all eight orientations and reflections of the image (positions, cell axes, image borders and Darcy initialisation transformed exactly). Without correction, the no-rotation model (three-seed mean) drops from 0.957/0.780/0.473 to 0.851/0.689/0.421 (dense 0.907/0.598/0.267 to 0.649/0.359/0.125), close to the rotation-augmented model (0.807/0.698/0.510 over the eight frames). The chamber axis can be estimated per sequence without labels from the sign of the nematic order $\langle \cos 2\phi \rangle$ of the cells' major axes (chosen on M0–M2; correct for both test sequences in all eight frames), leaving four axis-preserving frames. The rule fixed as primary before the test, which picks among these the frame with the highest model confidence, gives 0.942/0.764/0.465 (dense 0.873/0.556/0.236), recovering 90% and 80% of the no-rotation gain over the rotation-augmented model at 33 and 66 minutes; at 132 minutes there is no gain to recover over all windows. Averaging the predictions of the four frames, the rule we had planned first but replaced after a design check on the training movies, recovers all of it (0.962/0.813/0.504; dense 0.910/0.602/0.247). The orientation prior is therefore a property of the experiment that can be estimated from the detections, not knowledge supplied by the user.

E. Uncertainty, weighting and paired tests

Tab. S6 adds 95% bootstrap intervals over windows to the main table and weights the same predictions by targets. Weighting by targets lowers every method, most at long gaps, because dense windows are hardest; the ranking is unchanged. Window intervals are wide because windows differ in density, but paired comparisons over the same windows are tight. At 132 minutes the full model beats the family-centre regressor by 0.027 (95% CI 0.003–0.048; 25 of 33 windows), the ensemble with decoder by 0.071 (0.044–0.094; 29 of 33) and the annotation-free pipeline by 0.041 (0.018–0.062; 26 of 33). Ensembling and decoding improve the full model at every gap (+0.021/+0.083/+0.044, all intervals above 0.009), and annotation-free training costs 0.014/0.042/0.030 (intervals

Method	All windows 33	66	132	Dense (N_a>200) 33	66	132
Nearest ancestor	.497	.361	.263	.157	.083	.058
Radial expansion	.547	.458	.342	.249	.154	.056
Darcy flow prior (ours)	.627	.466	.368	.305	.161	.088
Optical flow (RAFT [8])	.636	.367	.244	.347	.094	.041
Unbalanced OT [7]	.555	.473	.353	.257	.159	.056
Non-rigid CPD [6]	.597	.466	.359	.294	.162	.060
Family-centre regression (GBDT)	.731	.610	.446	.449	.297	.201
Trackastra [4], greedy	.670	.239	.215	.366	.079	.062
Trackastra [4], argmax	.681	.176	.162	.395	.073	.057
Trackastra, fine-tuned per gap	.892	.716	.451	.788	.507	.171
Trackastra, fine-tuned over gaps	.864	.700	.488	.739	.476	.230
Trackastra, fine-tuned†	.892	.716	.488	.788	.507	.230
LineageRAFT, rot. aug.	.741	.604	.451	.459	.294	.178
LineageRAFT, no rot.	.853	.653	.447	.717	.443	.247
LineageRAFT, rot. aug. + Darcy init	.795	.692	.511	.557	.397	.222
Direct matcher (LineageRAFT, fixed centre)	.856	.684	.456	.706	.429	.207
LineageRAFT, full	.957	.780	.473	.907	.598	.267
LineageRAFT, full recipe, M0/M1 only, 20k steps	.715	.554	.394	.501	.289	.154
LineageRAFT, full + genealogy decoder	.966	.814	.485	.928	.650	.275
LineageRAFT, 3-seed ensemble	.964	.807	.490	.923	.634	.284
LineageRAFT, ensemble + decoder	**.977**	**.863**	**.517**	**.946**	**.703**	**.300**
LineageRAFT, ensemble + decoder, no annotation	.963	.822	.487	.919	.634	.260
Oracle family partition	.984	.929	.784	.963	.817	.465

Table S4. **In-domain results on OIAM M3/M4** (trained on M0–M2). Accuracy at gaps of 33/66/132 minutes. Trackastra receives images and ground-truth masks of both snapshots. †Fine-tuned on M0/M1 lineage, one model per gap or one over five gaps, whichever is better on M2 (per gap at 33/66, over gaps at 132 minutes; checkpoint and decoding also chosen on M2). M3/M4 were inspected during development (leave-one-movie-out: supplement). Direct matcher: the same network with the family centre frozen at its initial position. Rot. aug./no rot.: 30k steps, radial initialisation, with/without rotation augmentation; full: see text. Direct matcher and full (and its decoder row): mean of 3 seeds (std $\leq$0.016); ensemble rows average the seeds' probabilities; no annotation: both networks trained on self-labelled lineage of the unannotated M0–M2; oracle: ensemble; other learned rows: one seed.

	Family-centre regression			LineageRAFT, no rot., 30k			LineageRAFT, full		
Held out	33	66	132	33	66	132	33	66	132
M0	.724	.582	.408	.884	.697	.436	**.955**	**.748**	**.447**
M1	.777	.650	.451	.900	.721	.487	**.955**	**.824**	**.518**
M2	.698	.556	.409	.834	.631	.440	**.928**	**.764**	**.451**
M3	.738	.599	.434	.859	.630	.422	**.964**	**.793**	**.489**
M4	.724	.622	.459	.849	.690	.453	**.966**	**.830**	**.511**
Mean	.732	.602	.432	.865	.674	.447	**.954**	**.792**	**.483**

Table S5. **Leave-one-movie-out on OIAM.** Each row trains on the other four movies and tests on all windows of the held-out one. LineageRAFT, full: the configuration of the main results (80k steps, Darcy initialisation, consistency term, no rotation), one seed, fixed before running (protocol in the ledger). The five movies come from one experiment, so folds measure sensitivity to the movie, not independent replication. Bold: the best model per gap.

0.009–0.019, 0.029–0.054, 0.012–0.047).

Method	Window-averaged (95% CI) 33	66	132	Cell-weighted 33	66	132
Nearest ancestor	.497±.104	.361±.092	.263±.070	.183	.113	.098
Unbalanced OT	.555±.093	.473±.090	.353±.079	.244	.178	.138
Non-rigid CPD	.597±.093	.466±.091	.359±.076	.268	.179	.143
Trackastra, fine-tuned per gap	.892±.037	.716±.064	.451±.073	.762	.502	.243
Family-centre regression (GBDT)	.731±.083	.610±.092	.446±.075	.440	.321	.255
LineageRAFT, full (mean of 3 seeds)	.957±.019	.780±.053	.473±.060	.895	.599	.317
LineageRAFT, ensemble + decoder	.977±.014	.863±.050	.517±.062	.937	.698	.353
LineageRAFT, no annotation	.963±.018	.822±.053	.487±.070	.908	.636	.311
Oracle family partition	.984±.011	.929±.039	.784±.080	.954	.804	.560

Table S6. **Uncertainty and weighting** on OIAM M3/M4. Left: accuracy averaged over windows (as in the main table) with the half-width of a 95% bootstrap interval over windows. Right: the same predictions weighted by scored targets, which counts the dense windows more. Decoder rows use the windows with at least 9 targets.

Quality of the uncurated recA and rpsM lineages. Unlike FAST, recA and rpsM carry uncurated Schnitzcells lineages, so we audited them against FAST with statistics fixed in advance (`docs/schnitzcells_audit_protocol.md`), on exactly the scored targets. Neither contains non-binary divisions, interior orphans or lost cells. Size changes at division and large displacements are flagged often (76% and 9% of scored recA and rpsM targets lie on a flagged lineage path), but almost all flags are explained by one frame of growth (7.5 minutes) and whole-colony stage drift; after correcting for both (post hoc), 5.2% (recA) and 1.7% (rpsM) of scored targets lie on a flagged path, against 0.2% on FAST. Label errors on a fraction f of targets shift a method difference by at most $2f$; only the extreme case for recA (all flagged labels wrong, all in favour of the baseline) reaches its 0.10 margin. The margin over each colony's best geometric baseline is positive in all 25 colonies, unchanged in colonies with few flags, and of similar size on curated FAST, so label noise does not explain the zero-shot result; swaps of two adjacent cells of similar size would escape these checks.

F. Optimal-transport and registration baselines

Tab. S7 lists all six optimal-transport variants. Transport with uniform growth spreads every ancestor's mass over its neighbourhood: it fixes the most blatant failure of nearest assignment, interior ancestors receiving no target, but cannot know which ancestors grew or where the colony flowed, so in dense windows it stays below 0.30 even at 33 minutes, against 0.91 for LineageRAFT. Non-rigid CPD [6] treats ancestors as the centroids of a Gaussian mixture, moves them with a Gaussian-kernel displacement field of width β and regularisation λ (coordinates in ancestor spacings, 150 EM iterations), and assigns each target to the centroid

	All windows			Dense (N_a>200)		
Coupling, alignment	33	66	132	33	66	132
Exact capacity, none	.545	.506	.360	.217	.215	.079
Exact capacity, radial	.549	.476	.352	.251	.158	.056
Exact capacity, darcy	.560	.492	.372	.285	.197	.093
Entropic unbalanced, none	.561	.445	.337	.225	.112	.073
Entropic unbalanced, radial	.555	.473	.353	.257	.159	.056
Entropic unbalanced, darcy	.564	.491	.376	.298	.189	.102

Table S7. **Optimal-transport baselines** on OIAM M3/M4. Hyperparameters (capacity factor; entropy and KL relaxation) chosen on M0–M2. Costs are squared distances in units of the ancestor spacing.

of highest posterior. We searched $\beta \in \{2, 4, 8, 16, 32\}$, $\lambda \in \{1, 4, 16\}$, outlier weight $w \in \{0, 0.1\}$, with or without radial pre-alignment, on M0–M2; the selected setting (β=8, λ=4, w=0, radial) is interior to the grid, and its grid neighbours score within 0.007 on M0–M2. On M3/M4 it reaches 0.597/0.466/0.359 (dense 0.294/0.162/0.060), the level of unbalanced OT: a smooth deformation cannot follow the many independent family displacements of a dense colony.

Trackastra fine-tuned at each gap. The public `general_2d` model never saw gaps of an hour or more, so we also fine-tune it in domain, following a protocol fixed before training (one model per gap; architecture, features, loss, augmentation and optimiser of the official training script; batch 8 instead of 48 for memory; 20k steps). Training data are M0/M1 subsampled at the gap from every start frame, each detection's parent being its ground-truth ancestor one subsampled frame earlier, with images and ground-truth masks; the checkpoint (every 1k steps) and the decoding are chosen on M2, and M3/M4 are evaluated once. M2 accuracy rises from 0.66/0.16/0.17 off the shelf to 0.86/0.67/0.43; at 132 minutes it peaks at the first checkpoint and then declines while the training loss keeps falling, so that checkpoint is used. Argmax beats the greedy linker on M2 at every gap; on M3/M4 the greedy linker reaches 0.870/0.667/0.340 (dense 0.731/0.415/0.062) and argmax 0.892/0.716/0.451 (dense 0.788/0.507/0.171; main-text table at 33/66 minutes). Paired over windows, LineageRAFT (mean of three seeds) exceeds it by 0.065 [0.045, 0.087], 0.064 [0.043, 0.087] and 0.023 [−0.003, 0.048] at 33/66/132 minutes and wins every dense window at every gap; the ensemble with decoder exceeds it by 0.088/0.148/0.067, and the pipeline without annotation by 0.074/0.106/0.036 (all intervals above zero), winning every dense window at every gap. Fine-tuning closes most of the gap to LineageRAFT at 33 minutes but not in dense colonies (0.79/0.51/0.17 *vs*. 0.91/0.60/0.27), where many neighbours compete for each target.

Without rotation augmentation. Our in-domain models exploit the fixed chamber orientation by training without rotations, whereas the official Trackastra recipe rotates, flips and shears. Under a protocol fixed beforehand (`docs/trackastra_norot_protocol.md`) we fine-tuned Trackastra again with these three augmentations removed and everything else unchanged. It does not help: on M3/M4 it reaches 0.894/0.703/0.428 (dense 0.787/0.454/0.142) against 0.892/0.716/0.451, and it is worse on M2 at every gap (0.847/0.650/0.404 *vs*. 0.861/0.673/0.429), so the main-text row is unchanged. A first run that by mistake also doubled the batch to 16 gave 0.909/0.698/0.438; even the best Trackastra number at each gap, chosen on test, stays below LineageRAFT (0.957/0.780/0.473).

A learned tracker zero-shot. Of the bacterial datasets of the zero-shot table only FAST ships raw images (recA, rpsM and the blind-test subsets provide segmentations and lineage only). Under a protocol fixed before running (`docs/trackastra_zeroshot_fast_protocol.md`) we ran off-the-shelf Trackastra on every FAST window with the images and the *corrected* masks: 0.675/0.656/0.546 at 35/65/130 minutes with argmax decoding (greedy 0.688/0.624/0.553), against 0.789/0.697/0.640 for the positions-only RODSIM-trained LineageRAFT, which is better on all seven colonies (by 0.046–0.123).

Simulation or architecture? To separate what the simulator contributes from what family registration contributes zero-shot, we trained the direct matcher (the same network with the family centre frozen; main-text Table 2) on RODSIM exactly like the RODSIM-trained LineageRAFT, under a protocol fixed before training (`docs/sim_direct_matcher_protocol.md`). At the learning rate of LineageRAFT all three seeds diverged at the one-cycle peak (training loss rising from 1.0 to 30–1200); as fixed in an amendment before any evaluation, they were rerun at half the learning rate, and the diverged runs are also reported (0.05–0.48). Averaged over gaps and the three reruns, the direct matcher reaches 0.709/0.572/0.580/0.501 on FAST/recA/rpsM/OIAM against 0.709/0.560/0.564/0.520 for LineageRAFT (0.854/0.881 *vs*. 0.877/0.894 on T98G/LFCT), and it too beats all ten baselines on all 27 bacterial sequences. The pre-registered hypothesis that LineageRAFT beats it on all four bacterial datasets therefore **fails** (it holds on OIAM and, by less than 0.001, FAST); the hypothesis that the direct matcher beats the best baseline on at least three holds (4/4). The zero-shot transfer is the simulator's; the moving centre matters in domain and on the densest data

(OIAM), not on the small microcolonies of recA, rpsM and FAST, where LineageRAFT trails mostly at the longest gap (−0.03/−0.04).

Over several gaps, and on matched data. The in-domain comparison with fine-tuned Trackastra is asymmetric: Trackastra gets one model per gap and two training movies (M2 selects its checkpoint), LineageRAFT one model over horizons of 8–240 minutes and three movies, and Trackastra at 132 minutes peaks at its first checkpoint. Under a protocol fixed before training (`docs/matched_trackastra_protocol.md`) we (i) fine-tuned one Trackastra model on M0/M1 over gaps of 16, 33, 66, 132 and 240 minutes, keeping per evaluated gap the checkpoint best on M2, with the rule, fixed beforehand, that the main table takes whichever Trackastra variant is better on M2 at each gap; and (ii) trained the full LineageRAFT recipe on M0/M1 only for 20k steps (Trackastra's data and step budget; three seeds). Training over several gaps lowers Trackastra at 33/66 minutes (−0.028, −0.015 on M3/M4) but raises it at 132 minutes from 0.451 to 0.488 (+0.037 [0.017, 0.057]; dense 0.230), and it is also better there on M2 (0.466 *vs*. 0.429), so the main table now uses it at 132 minutes. Against it, the full model is ahead at 33/66 minutes (+0.093, +0.080, block CIs above zero) and behind at 132 minutes over windows (−0.015 [−0.044, 0.014]) but not weighted by cells (0.317 *vs*. 0.291); the ensemble with decoder leads by 0.029 [0.002, 0.058] (block CI −0.010–0.043) and the annotation-free pipeline ties (−0.001 [−0.032, 0.030]; −0.013 on M3, +0.012 on M4). Trained from scratch on M0/M1 only, LineageRAFT reaches 0.715/0.554/0.394 (dense 0.501/0.289/0.154; seeds within 0.012), far below both Trackastra variants: all three pre-registered hypotheses of the matched comparison **fail**. Its training accuracy (0.90) exceeds that of the full model at the same step (0.84), and the five movies have similar colony sizes, so it overfits two movies; Trackastra starts from weights pretrained on many datasets. LineageRAFT trained from scratch needs at least three annotated movies (or the annotation-free routes of the main text).

Optical flow. Under a protocol fixed before running (`docs/flow_baseline_protocol.md`) we computed pretrained torchvision RAFT flow [8] (RAFT-large C+T+S+K+H and RAFT-small C+T, 12 or 32 updates) between the two snapshot images, stretched to their 1st–99.5th intensity percentiles, and either traced targets back along the backward flow to the nearest ancestor or advected ancestors along the forward flow. Twice-upsampled inputs, also planned, were dropped before any test window was scored: RAFT's field on them was about 30 times larger than at native resolution in a three-window check on M0, and the all-pairs correlation did not fit on a shared GPU.

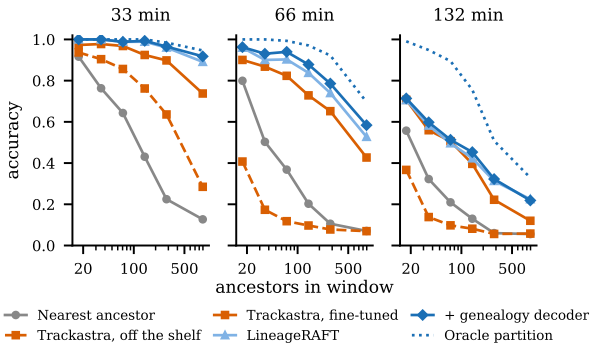


Figure S2. **Accuracy against colony size** on OIAM M3/M4 (windows binned by number of ancestors). Frame-to-frame methods collapse in dense colonies; family registration degrades gracefully, and the remaining gap to the oracle partition at long gaps is a grouping problem.

On M0–M2 the eight settings range from 0.372 to 0.413 (mean over gaps; nearest 0.368); the selected RAFT-small, forward, 12 updates reaches 0.636/0.367/0.244 on M3/M4 (dense 0.347/0.094/0.041), above nearest assignment at 33 minutes (+0.139, CI 0.098–0.182) but not at 66 or 132, and 0.23–0.41 below LineageRAFT at every gap. Flow trained on rigid scenes follows local motion over one cell cycle but cannot follow cells through several divisions in low-texture phase contrast.

Runtime. On the M3/M4 windows (one shared H200; median per window over sparse/dense windows, with a median of 32/527 ancestors and 72/1033 targets), Darcy initialisation takes 9/62 ms, one LineageRAFT pass 35/90 ms, the genealogy network 23/219 ms and our NumPy Potts decoder 3/574 ms; unbalanced OT takes 55/623 ms and exact capacity OT 0.3/57 ms. The largest window (1359 ancestors, 1485 targets) takes 3.4 s end to end, mostly in the decoder, which runs on the CPU over dense probabilities and could be restricted to candidate ancestors.

G. Accuracy against colony size

Fig. S2 breaks OIAM accuracy down by colony size. Nearest assignment and off-the-shelf Trackastra collapse as the colony grows, because more neighbours compete for each target and growth displaces cells further; fine-tuned Trackastra matches LineageRAFT in small colonies but falls behind beyond about 100 ancestors.

H. Biology read from two snapshots

Tab. S8 computes from predicted ancestry two readouts a biologist would take from lineage: size control (longer ancestors divide sooner) and the spread of family sizes across

Method	Size control (33 min) ρ	long	short	CV of family size 33	66	132	$r(\hat{n}, n)$ 66
Ground truth	.574	.789	.003	.369	.314	.496	1.000
Nearest ancestor	.134	.341	.211	.620	.919	1.567	.069
RODSIM only	.174	.395	.223	.412	.402	.357	.281
LineageRAFT, self-labelled	.489	.688	.039	.396	.441	.658	.679
LineageRAFT	.510	.711	.027	.397	.439	.646	.692

Table S8. **Biology read from two snapshots** (OIAM M3/M4, 32 random windows per gap and movie). Family size n is the number of descendants assigned to an ancestor. Size control: Spearman ρ between ancestor length (ranked within window) and n, and the fraction of the longest and shortest third of ancestors that divided within 33 minutes. CV: coefficient of variation of n across ancestors (growth heterogeneity). $r(\hat{n}, n)$: Pearson correlation of predicted with true family size at 66 minutes.

Model	Decoder	All 33	66	132	Dense 33	66	132
No rot.	argmax	.849	.653	.447	.717	.443	.247
	Potts β=1	.873	.688	.458	.763	.488	.248
	Potts β=2	.890	.706	.462	.786	.506	.240
	Potts β=4	.902	.729	.461	.799	.505	.229
	Cluster + argmax	.876	.724	.490	.769	.494	.253
	Oracle partition	.947	.855	.731	.876	.658	.380
Full	argmax	.961	.794	.486	.913	.603	.269
	Potts β=1	.966	.814	.493	.925	.635	.274
	Potts β=2	.969	.828	.498	.933	.656	.271
	Potts β=4	.973	.841	.506	.938	.661	.264
	Cluster + argmax	.955	.830	.526	.926	.649	.274
	Oracle partition	.982	.911	.764	.958	.777	.418

Table S9. **Partition-aware decoding** with single-snapshot genealogy on OIAM M3/M4. β was not tuned on held-out data; all values are shown; the Darcy-initialised model behaves alike. Cluster + argmax: average-linkage clusters of the genealogy affinities cut at 0.3, each cluster sent to the ancestor with the largest summed log-probability. Full is training seed 0; the main results table applies Potts β=2 to the probability ensemble of three seeds. Decoding skips the one 33-minute window with fewer than nine target cells, which moves 33-minute argmax values by up to 0.009 from other tables.

Gap	Accuracy	ECE	NLL of true ancestor
33 min	0.895 → 0.912	0.051 → 0.013	0.608 → 0.404
66 min	0.599 → 0.634	0.229 → 0.111	2.570 → 1.787
132 min	0.317 → 0.333	0.401 → 0.275	5.416 → 4.079

Table S10. **Calibration** of the full model on OIAM M3/M4, pooled over scored targets: mean of three single seeds → their probability-averaged ensemble. ECE: expected calibration error of the top choice (15 bins). Single models are overconfident; averaging at least halves ECE at 33 and 66 minutes.

ancestors, a measure of growth heterogeneity. Nearest assignment destroys the first and inflates the second 1.7- to 3.2-fold; both LineageRAFT models are close to the ground truth at 33 minutes, and at 66 minutes keep a correlation of 0.7 between predicted and true family size. Against the strongest baseline, under a protocol fixed before computing (`docs/offspring_readout_protocol.md`; decoder windows, scored targets only): assignment errors pool cells into the wrong families, so every method except a size-constrained decoder *overestimates* growth heterogeneity. The decoded LineageRAFT ensemble halves the inflation of decoded fine-tuned Trackastra (relative CV error 0.03/0.37/0.56 *vs*. 0.10/0.66/1.12 at 33/66/132 minutes; differences −0.29 [−0.36, −0.22] and −0.56 [−0.67, −0.45] at 66/132) and ranks ancestors by family size better (Spearman 0.96/0.66/0.48 *vs*. 0.87/0.48/0.35, CIs >0 at every gap); the annotation-free ensemble does likewise. Two caveats: at 132 minutes the estimated spread is still 55% too high, so quantitative heterogeneity needs gaps of at most an hour; and the GBDT regressor with its own size-constrained decoder estimates the spread best (0.02/0.21/0.31) while ranking ancestors worst among learned methods (Spearman 0.56 at 33 minutes).

I. Partition-aware decoding

Tab. S9 gives every decoder setting for two base models. Larger β helps at short and medium gaps but not on dense windows at 132 minutes, and hard clustering helps most at 132 minutes; β=2 is the setting reported in the main text. Both networks are trained on M0–M2, so no annotated movie is left to select it on; instead we cross-fitted the choice among $\beta \in \{1, 2, 4\}$ and Cluster + argmax, selecting on M3 and scoring M4 and vice versa. This gives equal or higher accuracy than fixed β=2 for every base model at every gap (three-seed ensemble 0.978/0.869/0.531 *vs*. 0.977/0.863/0.517; annotation-free pipeline without the consistency term, post hoc, 0.968/0.854/0.518 *vs*. 0.966/0.847/0.497), so the reported setting is not tuned to the test windows. Averaging the probabilities of the three training seeds helps the decoder more than any single model (0.057 *vs*. 0.034 at 66 minutes). Tab. S10 suggests why: single models are overconfident (mean confidence 0.83 at 66 minutes for an accuracy of 0.60), and the ensemble's probabilities are better calibrated, so the genealogy prior can overrule confident mistakes. Is calibration the reason? We fitted one temperature per model and gap by the likelihood of the true ancestor on M0–M2 (protocol fixed before running) and decoded M3/M4 once (Tab. S11). Even on the movies they were trained on, the models are overconfident (fitted T=1.6–1.8 for single seeds, 1.3–1.4 for the ensemble). On M3/M4 the temperature cuts the single models' ECE at 66 min-

	All windows			Dense		
Probabilities → decoder	33	66	132	33	66	132
Single seed, argmax	.955	.780	.473	.907	.598	.267
Single seed	.966	.814	.485	.928	.650	.275
Single seed, fitted T	.969	.826	.489	.934	.660	.269
Ensemble, argmax	.964	.807	.490	.923	.634	.284
Ensemble	.976	.863	.516	.946	.703	.300
Ensemble, fitted T	.977	.867	.515	.947	.705	.296

Table S11. **Temperature before decoding** (OIAM M3/M4; Potts β=2 unless argmax). T is chosen per model and gap by the likelihood of the true ancestor on M0–M2 and applied once. Single-seed rows: mean of three seeds. Here each seed is normalised before averaging, which moves ensemble rows by at most 0.001 from the main table. Decoding skips the one 33-minute window with fewer than nine target cells, which moves 33-minute argmax values by up to 0.009 from other tables.

		All windows			Dense (>200)		
Base	Decoding	33	66	132	33	66	132
Trackastra, fine-tuned	argmax	.889	.716	.451	.788	.507	.171
	+ decoder, T fitted on M2	.937	.785	.412	.869	.586	.151
	+ decoder, T=1	.939	.789	.475	.873	.601	.169
Family-centre regression	own geometric decoder	.717	.602	.441	.452	.289	.201
	argmax	.602	.494	.380	.390	.263	.168
	+ decoder	.521	.485	.359	.376	.261	.150
LineageRAFT, full (3 seeds)	argmax	.955	.780	.473	.907	.598	.267
	+ decoder	.966	.814	.485	.928	.650	.275
LineageRAFT, 3-seed ensemble	+ decoder	**.977**	**.863**	**.517**	**.946**	**.703**	**.300**

Table S12. **The genealogy decoder on baseline probabilities** (OIAM M3/M4, decoder windows; Potts β=2 on the same image affinities; protocol fixed before running). Trackastra: row-normalised association weights; GBDT: softmax of its own cost at its decoder temperature, and its own collective decoder with geometric affinities and capacity (the main-table setting).

utes from 0.23 to 0.14 and raises their decoded accuracy at that gap with every seed (+0.010 to +0.014), recovering about half of the gap between the single-model and ensemble decoder gains (0.034 and 0.057 at 66 minutes). It leaves the ensemble unchanged (within 0.004) and does not help dense windows at 132 minutes, so calibration explains part of the ensemble's advantage; the rest comes from its better base predictions. Seed averaging does not change the zero-shot picture: for the three RODSIM training seeds on the six datasets of the main transfer table, the ensemble is within -0.004 to $+0.011$ of seed 0 and 0.003–0.015 above the seed mean, and it beats the best geometric baseline on exactly the same sequences.

Is the decoder specific to LineageRAFT? No. Under a protocol fixed before running (`docs/decoder_on_baselines_protocol.md`) we applied the same decoder to fine-tuned Trackastra and to family-centre regression (Tab. S12). It lifts Trackastra by 0.05/0.07 at 33/66 minutes; at 132 minutes it helps with Trackastra's raw probabilities (+0.024) but hurts with the temperature fitted on M2 (-0.039), the pre-registered primary setting. On GBDT's unaries it is worse than GBDT's own collective decoder, which already couples neighbours through learned geometric affinities. All three hypotheses hold: decoded LineageRAFT beats decoded Trackastra at every gap (+0.029 [0.012, 0.049], +0.029 [0.006, 0.057], +0.073 [0.042, 0.103]) and decoded GBDT, and at 132 minutes the decoded ensemble beats both (+0.105 [0.077, 0.135] over Trackastra). Against the better-decoding raw-probability variant the margins are +0.027/+0.025/+0.010, the last with a CI including zero (-0.024–0.044); the decoded ensemble leads by +0.042 [0.007, 0.078] and the annotation-free ensemble by +0.024/+0.032/+0.012 (CIs above zero at 33/66). The long-gap advantage is therefore shared between the family-registration probabilities and the decoder, and at 132 minutes it needs the ensemble.

Overlapping windows. Test windows start every 33 minutes and span up to 132, so neighbouring windows share frames and window-level CIs are optimistic. Under a protocol fixed before computing (`docs/block_bootstrap_protocol.md`) we resampled blocks of four consecutive windows within each movie (Tab. S13). The pre-registered hypothesis, that every comparison significant over windows stays significant over blocks, **fails**: 14 of 19 do. Every margin at 33 and 66 minutes over fine-tuned Trackastra, the direct matcher and the regressor survives, as does the decoded ensemble's lead over Trackastra at 132 minutes. No undecoded single-model margin at 132 minutes survives (lower bounds -0.005 to -0.002), nor do the decoder-on-decoder margins at 66 minutes; the annotation-free pipeline's lead at 132 minutes is positive in both movies (+0.022, +0.051) but its block CI touches zero. Blocks of eight windows give the same picture.

Grouping, scored directly. Ancestor accuracy mixes two errors: splitting or merging families, and naming the wrong ancestor for a correct family. Tab. S14 scores grouping alone. The decoder's effect is mainly on grouping: for single models it raises pair family F1 by 0.05/0.10/0.04 but accuracy by only 0.01/0.03/0.01, and for the ensemble by 0.06/0.15/0.08 *vs*. 0.01/0.06/0.03; on dense windows at 66 minutes single-model F1 rises from 0.41 to 0.57. At 132 minutes a single LineageRAFT model groups barely better than the GBDT regressor (0.32 *vs*. 0.31) although it is more accurate; the decoded ensemble reaches 0.41.

What limits the decoder? In a post hoc diagnostic on the three-seed ensemble we replaced the predicted affinities by

Comparison	33 min			66 min			132 min		
	mean	windows	blocks of 4	mean	windows	blocks of 4	mean	windows	blocks of 4
Full − Trackastra, fine-tuned	+0.065	**+0.045, +0.087**	**+0.030, +0.087**	+0.064	**+0.043, +0.087**	**+0.037, +0.085**	+0.023	-0.003, +0.048	-0.019, +0.046
Full − direct matcher	+0.101	**+0.068, +0.138**	**+0.041, +0.138**	+0.096	**+0.072, +0.123**	**+0.057, +0.123**	+0.017	**+0.004, +0.031**	-0.005, +0.028
Full − regressor (GBDT)	+0.226	**+0.160, +0.297**	**+0.104, +0.331**	+0.170	**+0.126, +0.214**	**+0.095, +0.249**	+0.027	**+0.003, +0.048**	-0.002, +0.057
Ens. + dec. − Trackastra, fine-tuned	+0.088	**+0.063, +0.116**	**+0.041, +0.117**	+0.148	**+0.123, +0.174**	**+0.113, +0.183**	+0.067	**+0.041, +0.093**	**+0.032, +0.089**
No annotation − Trackastra, fine-tuned	+0.074	**+0.052, +0.098**	**+0.034, +0.097**	+0.106	**+0.081, +0.131**	**+0.074, +0.135**	+0.036	**+0.010, +0.064**	-0.001, +0.054
Full + dec. − Trackastra + dec. (fitted T)	+0.029	**+0.012, +0.049**	**+0.005, +0.041**	+0.029	**+0.006, +0.057**	-0.005, +0.049	+0.073	**+0.042, +0.103**	**+0.034, +0.106**
Full + dec. − Trackastra + dec. (T=1)	+0.027	**+0.011, +0.047**	**+0.005, +0.039**	+0.025	**+0.002, +0.051**	-0.007, +0.043	+0.010	-0.024, +0.044	-0.043, +0.051

Table S13. **Window *vs*. time-block bootstrap** (OIAM M3/M4; 95% CIs of the mean paired difference in accuracy, bold when above zero). Blocks: four consecutive windows (132 minutes of anchor times) resampled within each movie. Full: mean of three seeds; ensemble, decoder and no-annotation rows on the decoder windows.

Method	Accuracy			Pair family F1			F1, dense		
	33	66	132	33	66	132	33	66	132
Nearest ancestor	.484	.361	.263	.277	.238	.209	.078	.091	.102
Family-centre regression (GBDT)	.717	.602	.441	.541	.459	.309	.265	.192	.190
LineageRAFT (mean of three seeds)	.955	.780	.473	.884	.621	.319	.774	.408	.228
+ genealogy decoder	.966	.814	.485	.933	.717	.363	.871	.566	.282
LineageRAFT, 3-seed ensemble	.964	.807	.490	.906	.662	.330	.815	.449	.242
+ genealogy decoder	.977	.863	.517	.962	.812	.405	.919	.648	.295
+ decoder, no annotation	.963	.822	.487	.941	.756	.394	.881	.577	.262

Table S14. **Grouping, scored directly** (OIAM M3/M4, windows with at least nine targets). Pair family F1 counts pairs of scored targets that share the true ancestor and are given the same ancestor, whichever it is; the adjusted Rand index agrees within 0.03. The oracle partition scores 1 by construction. GBDT is retrained on M0–M2 here, so its accuracy differs slightly from the main table.

Simulator variant	FAST	recA	rpsM	OIAM
Full RODSIM (256 colonies)	.712	.566	.571	.525
No physics randomisation	.701	.552	.562	.521
No field-of-view crop	.696	.549	.559	.507
64 colonies	.713	.560	.577	.524
16 colonies	.720	.580	.592	.522
+ division modes	.658	.507	.504	.485

Table S15. **Simulator ablations** (zero-shot accuracy, mean over gaps; every row is one training seed, seed 0, so it differs from the three-seed means of the zero-shot table).

true ones (1 if two neighbours share an ancestor H minutes earlier, 0 otherwise) and widened the genealogy graph from 8 to 16 or 24 neighbours. With predicted affinities the wider graph changes accuracy by at most 0.004 at any gap. With true affinities it lifts 132-minute accuracy from 0.61 to 0.65 (β=2) and 0.70 (β=4, 24 neighbours), still below the 0.78 of the oracle partition (dense windows: at most 0.40 *vs*. 0.47). Potts mean field, however, is a weak partitioner even with true affinities. In a follow-up fixed in advance (`docs/global_partition_protocol.md`) we replaced it by non-local partitioners on 8- and 24-neighbour target graphs: correlation clustering of the affinity log-odds (greedy additive edge contraction with local moves), the same objective with LineageRAFT' co-assignment log-odds added, and spectral clustering with as many clusters as distinct argmax ancestors, each partition assigned to ancestors by Cluster + argmax or Hungarian matching. With predicted affinities none improves on Potts at 132 minutes (best 0.517 *vs*. 0.517, paired CI −0.014–0.014; dense 0.293 *vs*. 0.300; spectral 0.45), although pair family F1 rises from 0.41 to 0.47; at 66 minutes all are below Potts. With true affinities, correlation clustering on the 24-neighbour graph with Hungarian assignment reaches 0.780 (dense 0.454), the oracle partition's 0.784 (0.465). The remaining gap at long gaps is therefore the accuracy of single-image genealogy, not the locality of the decoder.

J. Simulator ablation

Tab. S15 retrains the positions-only RODSIM model with one ingredient changed and evaluates it zero-shot. Removing physics randomisation or the field-of-view crop costs at most 0.02, and 16 simulated colonies are as good as 256, whereas randomising division modes (asymmetry, V-snapping) hurts transfer of the assignment model by about 0.05 (it helps the genealogy model, Appendix K). What the network extracts from RODSIM is a simple prior on how growing rods push each other apart.

K. A pipeline without any real label

Tab. S16 decodes the RODSIM-trained LineageRAFT with RODSIM-trained genealogy. Genealogy with division modes improves every gap (up to 3 points), the symmetric simulator's does not; genealogy trained on real labels gains about three times as much. Real images need not mean real labels: the genealogy network trained on chained pseudo-lineage of the unannotated M0–M2 movies (same links as the self-labelled assignment model; protocol fixed before training) agrees with the annotated pair labels on 94–100% of pairs in sampled M0 frames (lowest for dense frames and long horizons) and reaches 0.96/0.89/0.69 AP on M3/M4 at 33/66/132 minutes (annotated: 0.97/0.91/0.70). Post hoc, the self-labelled assignment model without the con-

Genealogy in the decoder	33	66	132
none (argmax)	.656	.516	.393
RODSIM	.659	.520	.376
RODSIM + division modes	.667	.546	.408
OIAM M0–M2 labels	.696	.597	.434
oracle partition	.768	.718	.679

Table S16. **A label-free pipeline.** LineageRAFT trained only on RODSIM, decoded with single-image genealogy from different sources (Potts, β=1); OIAM M3/M4 accuracy. The first three rows use no real label. Decoding skips the one 33-minute window with fewer than nine target cells, which moves 33-minute argmax values by up to 0.009 from other tables.

sistency term (three seeds), decoded in the same way, reaches 0.966/0.847/0.497. In the Potts decoder (β=2) it changes the accuracy of each full-model seed by at most 0.005 and of the three-seed ensemble by at most 0.003 (dense: 0.004) (0.976/0.861/0.520 *vs*. 0.977/0.863/0.517; dense 0.945/0.699/0.296 *vs*. 0.946/0.703/0.300).

L. Robustness to detection errors

Fig. S3 perturbs both snapshots of every OIAM M3/M4 window at the 33-minute gap with position noise, missed detections, merges of a cell with its nearest neighbour (under-segmentation) and spurious detections; only targets that survive unmerged and whose ancestor survives unmerged are scored. With 10% of cells missed, merged or hallucinated, LineageRAFT is still correct for 90–91% of targets, against about 50% for nearest assignment on the same targets. Position noise of 0.3 cell spacings, far beyond the localisation error of a modern segmenter, lowers accuracy to 0.82. The ordering is unchanged at 66 and 132 minutes. With Cellpose-SAM masks, the loss concentrates in the first, sparse frames, where a few spurious objects distort the population-size normalisation.

A second segmenter. Tab. S17 adds Omnipose [3], run with the settings of its official bacterial phase-contrast notebook under a protocol fixed before any test frame was segmented (one amendment: a dependency pin needed to import it). It finds every annotated cell but adds 16–43 small objects to every M3 frame, out-of-colony debris that Cellpose-SAM mostly ignores. On M4 the full model keeps 0.94/0.74/0.46 with Omnipose masks; on the small early M3 colonies the debris corrupts the population-size inputs (N_a, N_b, centroid, spacing) and accuracy falls below nearest assignment. Dense windows lose at most 0.08. We did not filter the debris, since any filter would be chosen on the test movies; a size or colony-membership filter is the obvious remedy and the population-size inputs are the component to harden.

M. Self-labelling details

Linking. One-frame links are solved with `scipy`'s exact linear assignment over the 8 nearest slots of each target. On OIAM (masks available) the daughter slots sit at $\pm L/4$ along the mother's axis, with $L = \sqrt{12}$ times the major-axis standard deviation of the mask; the extra division cost is 10 px and the no-parent cost 60 px (0.55 and 3.5 median spacings). For the external sources, which provide positions only, both daughter slots sit at the mother's centre and costs are scaled by the median spacing of each frame. These values were set on OIAM M0 and never tuned on the external sources.

Does self-labelling need family registration? Under a protocol fixed before training (`docs/pseudo_direct_matcher_protocol.md`) we trained the direct matcher (frozen family centre) on the same pseudo-lineage, recipe and seeds as self-labelled LineageRAFT. Both pre-registered hypotheses hold. Self-labelling is not specific to LineageRAFT: it costs the direct matcher only 0.009/0.011/0.012 (0.847/0.673/0.444 *vs*. 0.856/0.684/0.456 on true lineage). The architecture margin survives it: self-labelled LineageRAFT leads the self-labelled direct matcher by 0.088 [0.058, 0.122] and 0.059 [0.037, 0.082] at 33/66 minutes (0.008 [−0.005, 0.021] at 132; dense 0.865/0.533/0.244 *vs*. 0.686/0.412/0.199), and even the direct matcher trained on true lineage by 0.079 and 0.048 (CIs above zero). In domain, family registration with no annotation thus beats cell matching with annotation at 33 and 66 minutes.

Quality of pseudo-lineage. Tab. S18 reports the fraction of cells whose chained ancestor equals the annotated one. Errors at divisions dominate without the fission prior: on M0, one-frame links are wrong for 30% of dividing mothers and 0.5% of the others; the daughter slots reduce the former to 4%. Because a swap between two sisters does not change ancestry at longer gaps, the composed accuracy decays slowly with H.

Protocol 3. The protocol (`docs/blind_test3_protocol.md`) fixed, before training: two folds per source by sorted sequence name; 6k fine-tuning steps from the RODSIM model at learning rate 2×10^{-4}, positions only, horizons log-uniform in 15–240 minutes; evaluation on the windows of blind tests 1 and 2 with the same seeds. One amendment was made before any evaluation: some short *B. subtilis* sequences admit no frame pair at long horizons, so the sampler draws another sequence when 100 draws fail. Hypotheses and outcomes: A1 (SV with the adapted model beats the best of eight geometric baselines) holds on TrackRefiner, TrackMate

Model	Positions	Detection Rec.	Detection Prec.	All windows 33	All windows 66	All windows 132	Dense (N_a>200) 33	Dense (N_a>200) 66	Dense (N_a>200) 132
LineageRAFT	true	1	1	.957	.780	.473	.907	.598	.267
	Cellpose-SAM	1.000	.936	.922	.730	.436	.902	.592	.262
	Omnipose	1.000	.852	.736	.503	.287	.877	.522	.220
LineageRAFT, no annotation	true			.935	.733	.452	.865	.533	.244
	Cellpose-SAM			.918	.697	.420	.859	.531	.243
	Omnipose			.749	.500	.281	.843	.489	.207
LineageRAFT, RODSIM only	true			.657	.512	.390	.333	.197	.096
	Cellpose-SAM			.562	.438	.342	.342	.201	.099
	Omnipose			.486	.355	.230	.332	.195	.104
Nearest ancestor	true			.497	.361	.263	.157	.083	.058
	Cellpose-SAM			.497	.359	.261	.155	.081	.055
	Omnipose			.495	.360	.259	.156	.080	.047

Table S17. **End-to-end with two off-the-shelf segmenters** on OIAM M3/M4, neither fine-tuned nor tuned (Cellpose-SAM; Omnipose `bact_phase_omni` with the settings of its official notebook, fixed before segmenting). Accuracy is correct over *all* annotated targets (missed or mismatched cells count as errors), mean of three training seeds. Both segmenters find essentially every cell; losses come from spurious objects, under Omnipose mostly out-of-colony debris in the sparse early frames of M3, so dense windows lose at most 0.08.

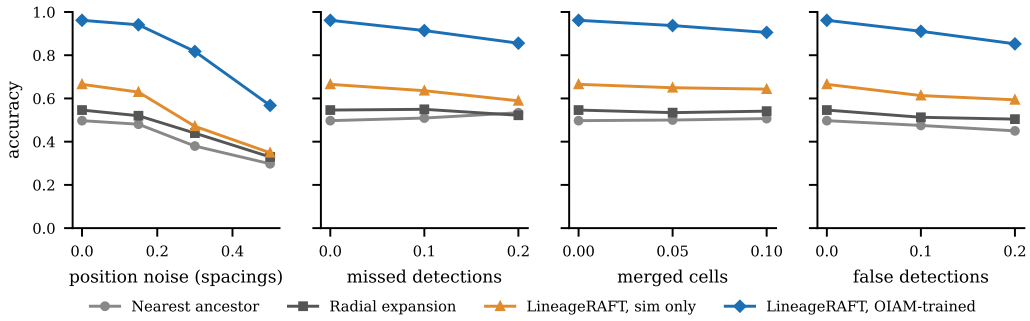


Figure S3. **Robustness to detection errors** on OIAM M3/M4 at a 33-minute gap. Each perturbation is applied independently to both snapshots; the leftmost point of every panel is the unperturbed result.

Data (gap)	33/30	66/60	132/120	240
OIAM M0, fission prior	.977	.930	.921	.927
OIAM M0, no daughter slots	.844	.700	.570	.522
OIAM M1	.976	.930	.929	.928
OIAM M2	.910	.912	.893	.881
B. subtilis (positions only)	.976	.968	.927	–
LYN yeast	.950	.914	.852	–
TrackMate	.892	.834	.869	–
TrackRefiner	.739	.654	.551	–

Table S18. **Accuracy of chained pseudo-lineage** against the annotation, weighted by cells, over random windows (minutes; OIAM at 33/66/132/240, external sources at 30/60/120). TrackRefiner movies are sampled every 2 minutes with large motion between frames.

and LYN and fails on *B. subtilis* (0.894 *vs*. 0.895); A2 (it beats SV without adaptation) holds on TrackRefiner, LYN and *B. subtilis* and fails on TrackMate (0.803 *vs*. 0.838); A3 (the adapted whole-field model beats the RODSIM model) holds on all four.

Protocols 4 and 5. On FAST, recA and rpsM (`docs/selflabel_ext_protocol.md`), the same procedure with positions-only Hungarian links failed: adaptation was below the RODSIM model on every dataset and all 25 colonies (Tab. S19). Measured afterwards against the ground truth, one-frame Hungarian links are only 0.55 (recA), 0.84 (rpsM) and 0.94 (FAST) correct at these frame rates, and compose to 0.24–0.68, below the RODSIM model's direct accuracy; one-frame argmax links of the RODSIM model itself are 0.73, 0.94 and 0.99 correct. Protocol

Dataset	Best geom.	RODSIM	+ Hungarian	+ bootstrap	wins
FAST	.593	.702	.642	**.730**	6/7
recA	.466	.563	.445	**.576**	6/9
rpsM	.455	.560	.450	**.604**	9/9

Table S19. **Temporal bootstrapping** on the microcolony datasets (accuracy, mean over colonies of the mean over gaps; two folds per dataset, each model evaluated on the other fold). The RODSIM model is fine-tuned on pseudo-lineage chained from positions-only Hungarian links (protocol 4) or from its own one-frame predictions (protocol 5). Wins: colonies on which bootstrapping beats the RODSIM model.

5 (`docs/selflabel_bootstrap_protocol.md`), fixed before training and run once, changes only the one-frame links to the RODSIM model's predictions; it is post hoc in that the data had been scored. Both hypotheses hold on all three datasets: bootstrapping beats the RODSIM model (C1) and the best geometric baseline (C2).

Protocol 6. We then applied bootstrapping to the four blind sources of protocol 3 (`docs/selflabel_bootstrap_blind_protocol.md`; same folds, windows and settings), again fixed before training and run once. E1 (the bootstrapped whole-field model beats the Hungarian-adapted one) fails on all four sources: 0.405 *vs*. 0.478 on TrackRefiner, 0.549 *vs*. 0.704 on TrackMate, 0.846 *vs*. 0.849 on yeast and 0.885 *vs*. 0.887 on *B. subtilis*. E2 (SV with the bootstrapped model beats the best geometric baseline) holds on TrackRefiner (0.559 *vs*. 0.461), TrackMate (0.832 *vs*. 0.796) and yeast (0.858 *vs*. 0.827) and fails on *B. subtilis* (0.894 *vs*. 0.895). E3 (it beats SV with Hungarian adaptation) holds on TrackRefiner, TrackMate and yeast by 0.002–0.029 and fails on *B. subtilis* (0.8937 *vs*. 0.8942). A likely reason, tested below: on these fields of many small colonies the whole-field RODSIM model is itself poor (0.30 on TrackRefiner), so its one-frame links are a worse teacher than Hungarian links, whereas on FAST, recA and rpsM it was the stronger of the two. Bootstrapping is therefore useful when the teacher is reliable, which can be checked without labels only indirectly.

Can the teacher be judged without labels? A further protocol (`docs/teacher_reliability_protocol.md`), fixed before computing, measured six label-free statistics of the RODSIM teacher's one-frame links on every sequence that supplied pseudo-labels (114 sequences or colonies of the ten sources): mean top-1 probability, entropy, agreement with Hungarian links (primary), fraction of ancestors with descendants, agreement with nearest assignment, and a fission-aware cycle consistency. All six track the teacher's own link accuracy (Spearman $|\rho|$ 0.70–0.90; primary 0.84, 95% CI 0.74–0.90), but none decides whether bootstrapping beats Hungarian self-labelling: no single threshold on the source mean of the primary statistic separates the six sources where it won from the four where it lost (best: 2 of 10 wrong), and the same holds for the other five. With ground truth the rule is simple: bootstrapping won exactly where the teacher's one-frame links are more accurate than Hungarian links (all ten sources). On TrackRefiner and TrackMate they are not (0.78 *vs*. 0.96 and 0.83 *vs*. 0.98), confirming the explanation above; on yeast and *B. subtilis* both are 0.99. Agreement is symmetric and cannot tell which linker is wrong. A rule combining teacher confidence and agreement separates all ten sources but was found after seeing them and is untested.

N. Blind tests 1 and 2, per-colony processing and self-labelled adaptation

All datasets above were inspected during development, so we fixed a protocol (data, gaps, models, baselines, metric, hypotheses) before running anything on 11 unseen sequences: eight TrackRefiner ground-truth movies [1] and three curated TrackMate *E. coli* movies [2]. The single run failed (Tab. S20, left): the RODSIM-trained model loses to the best geometric baseline on both sources and wins on only 2 of 11 sequences, the only two that show a single colony. The other fields hold 8 to 70 separate microcolonies or are steady-state chambers that flush cells out, so descendants do not move radially as LineageRAFT assumes.

Per-colony processing and label-free selection. From this diagnosis, ancestors are split into colonies by single linkage at three cell spacings, each target joins the colony of its nearest ancestor, and every colony is a separate input. Since no single method is best everywhere (registration wins in crowded, expanding colonies, nearest assignment where cells barely move), we pick per input between nearest assignment and per-colony LineageRAFT using a biological consistency check: in a growing population every ancestor has at least one descendant, so we keep the hypothesis under which more anchors receive a target (ties: more uniform family sizes). This survival-based selection (LineageRAFT-SV) is within 0.005 of a per-window oracle choice of its components on the first blind set, post hoc.

Second blind test. A second protocol, fixed before downloading the data, tested SV on six budding-yeast colonies from LYN [10], a different kingdom with asymmetric division, and 44 mostly sparse *B. subtilis* spore-outgrowth sequences [5]. On yeast SV beats every geometric variant (0.842 *vs*. 0.827) but on only half of the colonies; on *B. subtilis*, where most spores neither move nor divide, it is 0.008

	Blind test 1		Blind test 2	
Method	TrackRefiner	TrackMate	LYN (yeast)	*B. subtilis*
Nearest ancestor	.407	.796	.827	.895
Best of 8 geometric	.461	.796	.827	.895
LineageRAFT, whole field	.306	.419	.817	.874
LineageRAFT, per colony	.501†	.661†	.828	.878
LineageRAFT-SV (label-free selection)	.496†	.838†	.842	.887
SV wins / sequences	6/8†	3/3†	3/6	9/44
+ self-labelling on the other fold's unannotated sequences (protocol 3)				
LineageRAFT, whole field	.478	.704	.849	.887
LineageRAFT, per colony	.560	.753	.856	.890
LineageRAFT-SV	.555	.803	.856	.894
SV wins / sequences	8/8	2/3	4/6	22/44

Table S20. **Pre-registered blind tests and self-labelled adaptation.** Accuracy averaged over windows, gaps (30/60/120 min) and sequences; RODSIM-trained, positions only. Test 1 (11 sequences) was run first and failed; †per-colony processing and SV were designed after seeing it, so these test-1 entries are post hoc (SV there uses one seed). Test 2 (6 yeast colonies, 44 *B. subtilis* sequences) was fixed before its data were downloaded and run once. Best of 8 geometric: nearest, centroid translation, radial expansion and radial sector (the ancestor in the most similar direction from the colony centroid), each on the whole field or per colony, best chosen per source after the run. Protocol 3 (not blind: same windows, method fixed before running) fine-tunes the RODSIM model on chained pseudo-lineage of the other half of each source's sequences, with no ground-truth lineage.

below nearest assignment (0.895), failing all three hypotheses.

Self-labelled adaptation. Before adapting, we fixed a third protocol on the same windows (not blind, since the data had been scored): within each source, sequences are split into two folds, the RODSIM model is fine-tuned for 6k steps on positions-only pseudo-lineage of one fold, and evaluated on the other. Adaptation improves the whole-field model on all four sources (Tab. S20, bottom), most where simulation failed: from 0.31 to 0.48 on TrackRefiner and from 0.42 to 0.70 on TrackMate. With SV it beats the best geometric baseline on TrackRefiner (0.555 *vs*. 0.461, all 8 sequences), TrackMate and yeast, but not on *B. subtilis* (0.894 *vs*. 0.895). On the microcolony datasets of the zero-shot table (FAST, recA, rpsM), Hungarian links are poor at their frame rates and adaptation hurts on all 25 colonies; with temporal bootstrapping, fixed in a further protocol, it beats the RODSIM model on all three datasets and 21 colonies, but not Hungarian self-labelling on the four blind sources (below).

O. Blind-test protocols

Table S21 lists every protocol fixed before running, with its outcome.

Blind test 2. The protocol (`docs/blind_test2_protocol.md`) was fixed before the LYN and *B. subtilis* data were downloaded and run once with the positions-only RODSIM model. Hypotheses, per source: H1, SV beats the best of eight geometric variants (chosen per source after the run); H2, it does so on most sequences; H3, SV is at least as accurate as both of its components (nearest assignment and per-colony LineageRAFT). On yeast H1 and H3 hold (0.842 *vs*. 0.827 and 0.828) and H2 fails (3 of 6 colonies); on *B. subtilis* all three fail (0.887 *vs*. 0.895; 9 of 44 sequences).

Blind test 4. The protocol (`docs/blind_test4_protocol.md`) was fixed before any file of the trpL, metA and pheA subsets of [9] was downloaded; only the archive listings, which give colony names, were read. It fixed all colonies with Schnitzcells lineage (10, 9 and 9), the 7.5-minute frame interval stated in that paper, gaps of 4, 8 and 16 frames, the recA window sampler and seed, six baselines (four geometric; exact and unbalanced OT without alignment at the OIAM-selected settings), the RODSIM model, and Hungarian and bootstrapped self-labelling with protocol-3 settings and folds by sorted colony rank. Hypotheses, per subset: F1, RODSIM beats the best baseline; F2, on more than half of the colonies; F3, bootstrapping beats RODSIM; F4, bootstrapping beats Hungarian self-labelling. One amendment was made before any window was scored: nine frames (in seven colonies) store an empty crop rectangle that the parser rejects; they are dropped and logged, and windows ending on them do not exist. No colony was excluded. All hypotheses hold on all subsets (main-text table), at every gap: RODSIM exceeds the best baseline by 0.11–0.14 at 30 minutes and 0.06–0.09 at 120 minutes, and bootstrapping adds a further 0.03–0.06 at every gap.

P. Where do long-gap errors go?

For every wrong target in 24 random windows per gap and movie of OIAM M3/M4 we measured how long before the first snapshot the lineages of the predicted and the true ancestor split. If errors at long gaps were confusions between sisters born just before the first snapshot, they would be an identifiability limit. They are not: at 132 minutes only 6% of LineageRAFT's errors confuse ancestors that split within 33 minutes, and 19% within 132 minutes; forgiving all confusions between ancestors that split within 66 minutes raises cell-weighted accuracy only from 0.33 to 0.40. The predicted ancestor is instead a neighbour of the true one, a median of 1.8 cell spacings away (nearest assignment: 3.6). The residual error at long gaps is therefore a registration error of one to two cells, which is consistent with the oracle-partition result: knowing which cells form a family places most families correctly. Feeding the predicted family centres back as the initialisation for a second pass does not refine them; it lowers accuracy at every gap (0.33 to 0.19 at

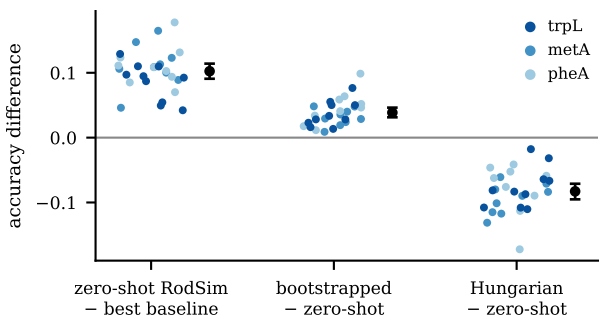


Figure S4. **Blind test 4, colony by colony** (28 colonies, accuracy averaged over the three gaps). Left: the zero-shot RODSIM model minus the best of the six baselines *for that colony*: positive on 28/28, mean $+0.102$ (95% bootstrap CI over colonies 0.091–0.114). Middle: bootstrapped self-labelling minus zero-shot: 28/28, $+0.039$ (0.031–0.046). Right: Hungarian self-labelling minus zero-shot: negative on 28/28, -0.083 (-0.095 to -0.071). Black: mean and CI.

132 minutes), because the network learned offsets relative to the Darcy initialisation.

Q. Further negative results

More refinement steps. Doubling the refinement rounds of the full model to $T{=}8$ (unshared weights, otherwise identical, 80k steps, one seed) lowers accuracy at every gap: 0.94/0.75/0.44 *vs*. 0.96/0.79/0.49 for $T{=}4$ (seed 0), and 0.88/0.58/0.24 *vs*. 0.91/0.60/0.27 on dense windows, although its final training accuracy is the same (0.96). Consistent with the long-gap analysis above, the residual error is not resolved by iterating the same local update longer.

Fission-prior decoding. Replacing argmax by an exact MAP assignment under a log-normal prior on family size around N_B/N_A (min-cost matching over ancestor slots with convex costs) changes accuracy by at most 0.01 (full model: 0.794→0.800 at 66 minutes). The mean family size is right, but its spread is not: Appendix R shows that predicted families are about twice as dispersed as true ones, and that a prior which repairs this still leaves long-gap accuracy unchanged.

Forward simulation. FSA initialises RODSIM with the first snapshot (doubling time $H \ln 2/\ln(N_B/N_A)$), simulates H minutes eight times, warps each rollout onto the second snapshot and assigns every target to the ancestor whose simulated descendants are densest there. FSA's alignment by a similarity transform gives 0.60/0.51/0.38 on OIAM; a smooth non-rigid warp raises it to 0.63/0.53/0.39 and dropping the similarity step to 0.68/0.55/0.40. These rollouts start from the true cell lengths and orientations; from positions only, as in the transfer experiments, FSA gives 0.57/0.45/0.36. With shapes (mean 0.54) it is the only baseline above the positions-only RODSIM model on OIAM (0.52); the other datasets lack comparable shape features. Selecting rollouts by Chamfer distance to the second snapshot, or calibrating the doubling time so that the simulated count *inside the field of view* matches N_B, does not help on dense windows (0.52/0.29/0.14 *vs*. 0.54/0.30/0.14).

R. Diagnosing long gaps under one protocol

Table 2 under one protocol. Table S22 recomputes every row of Tab. 2 of the main paper from its per-window results with the code that writes the table, on one window set: every value is reproduced exactly, and no row lacks a window except the decoded rows, which skip the one 33-minute window with fewer than nine targets. All rows use the same scorer, filter and test movies; the confidence intervals use a time-block bootstrap because consecutive windows overlap. At 132 minutes there are 33 windows (10 dense), so marginal intervals are about ± 0.09 over all windows; paired comparisons (Supp. I) are tighter because window difficulty cancels.

Which error is it? We split long-gap errors by intervention on the same ensemble scores (Tab. S23; protocol fixed before the run). Candidate retrieval is not the bottleneck: the true ancestor is among a target's 24 candidates for 95–98% of targets at 132 minutes, and only 7–9% of errors are outside them. Giving the true partition fixes more errors over all windows (0.49→0.79 at 132 minutes) than giving the true family centres (0.61), but on dense windows the two are equal (0.47 and 0.50) and strongly super-additive: together they reach 0.90. The largest class of dense 132-minute errors (28%) is fixed only when both are given; 20% are fixed by the centre alone, 21% by the partition alone, 14% by either. How precise must the centre be? Perturbing the true centres by isotropic noise of σ anchor spacings, with the true partition, keeps dense 132-minute accuracy at 0.89 up to $\sigma{=}0.5$, 0.79 at $\sigma{=}1$ and 0.44 at $\sigma{=}2$, while the network's centres are off by a median of 2.5 spacings there. Long gaps therefore need family centres within about one spacing *and* the partition, jointly; improving either alone is bounded near 0.5 on dense windows.

A structured decoder with a growth prior. At 132 minutes predicted families are about twice as dispersed in size as true ones (std 1.39 for argmax and 1.26 for Potts against 0.81 on M4), with 3–9 times too many single-cell families, and each true family is split over 2.1–2.6 ancestors. Binary fission makes true family sizes sub-Poissonian (Fano factor 0.16–0.28 in RODSIM), which a pairwise Potts energy cannot express. We therefore decoded

$E(\pi) = -\sum_j \log p_{j\pi(j)} - \lambda \sum_{(j,k)} [s_{jk} \log w_{jk} + (1-s_{jk}) \log(1-w_{jk})] + \mu \sum_i \phi_i(n_i)$, with $s_{jk}=[\pi(j)=\pi(k)]$ and ϕ_i a Gaussian prior on the visible family size whose mean follows from mass conservation and the family's visible fraction and whose variance combines the RODSIM Fano factor with binomial thinning at the image border (no real label). Inference runs mean field on the pairwise term and then the exact MAP of the convex cardinality term by min-cost matching over ancestor slots; one optional round re-scores candidates relative to the decoded family centroids. $\lambda=\mu=1$ were fixed by theory, and the protocol before M4 was decoded. On M4 (Tab. S24) the decoder repairs the failure it targets: at 132 minutes the size error falls by 0.20 [0.09, 0.32] (dense 0.52), singletons by 0.02, the size spread to within 20% of the truth, and pair F1 rises by 0.03 (66 minutes: 0.06–0.09). Accuracy rises at 66 minutes (+0.015 [0.007, 0.023]; dense +0.020) but not at 132 (-0.009 [-0.036, 0.016]), and the centre update gives nothing (H1, H2 hold; H3 fails at 132). Even the true family sizes reach only 0.526 (dense 0.312) at 132 minutes. On FAST/recA/rpsM the decoder lowers the size error of both RODSIM networks everywhere but also their accuracy (H4 fails). Decoding takes 0.6 s per window (dense 1.8 s), Potts 0.25 s. A size prior thus fixes how many cells a family has, not where it went: consistent with Tab. S23, grouping without registration is bounded at long gaps.

Where does the moving centre help? Trained on RODSIM, LineageRAFT and the direct matcher tie zero-shot (Supp. F), whereas on OIAM the centre adds 0.10 at 33/66 minutes. Table S25 conditions the margin on the window. The external bacterial windows hold a median of 4–6 ancestors, where even the OIAM-trained centre adds only 0.01, so these datasets cannot show a difference. The decisive row is OIAM itself: on windows of more than 400 ancestors the OIAM-trained centre adds 0.19, the RODSIM-trained one 0.016. In a regression on log ancestors, displacement, family mixing and generations, the margin grows by 0.037 per doubling of ancestors in-domain ($p<10^{-4}$) and by -0.0004 (n.s.) zero-shot. The training distributions differ exactly there: windows sampled as in training (1,500 per source) have more than 400 ancestors in 22% of OIAM samples and 10% of RODSIM ones, and in those the median displacement is 2.7 cell spacings on OIAM (90th percentile 7.6) against 1.6 (3.8) in RODSIM. RODSIM rarely shows crowded families that moved by several cells, which is the regime in which the centre pays off. Rarity is not the whole explanation: on 600 windows of 32 held-out RODSIM colonies the RODSIM-trained margin also grows with crowding (+0.004 per doubling, $p<10^{-4}$), but above 400 ancestors, at a median displacement of 2.0 spacings, it is only +0.024 [0.013, 0.036], a tenth of the in-domain margin. We therefore attribute zero-shot transfer to the simulator and in-domain registration gains to the moving centre, and do not claim that the centre is what transfers.

Three attempts at long gaps. Table S26 reports three follow-ups to the diagnosis above, each run once under hypotheses fixed beforehand (files in `docs/`). *Coupled refinement.* Since centre and partition errors are super-additive, we let ancestors compete for targets inside every refinement round: an MLP scores each ancestor's claim on its 24 nearest targets, claims are normalised over all ancestors claiming the same target (E-step), and the resulting soft family size and claimed centroid enter the ancestor state, through a zero-initialised layer, before the centre update (M-step). The model equals LineageRAFT at initialisation and uses no new loss. On a development split (trained on M0/M1, validated on M2, three seeds per arm) it gained +0.052/+0.039/+0.018 and was adopted; trained with the full recipe and tested on M3/M4, it does not improve accuracy over all windows (C1 fails), but it does on dense windows at 66 and 132 minutes (C2), with a lower family-centre error (2.36 *vs.* 2.43 spacings, median over dense 132-minute targets; C3) and a higher pair F1 (C4, CI includes 0). The development gain did not survive training on three movies for four times longer, and the centre error remains far from the one spacing that the oracle sweep requires. Two further levers on the coupled model were rejected on the development split by the same rule: eight refinement rounds instead of four (-0.058/-0.038/-0.016; +0.012 on dense 132-minute windows) and training only on gaps of 33–240 minutes (-0.059/-0.032/-0.016), so short gaps in training help long gaps too. *A simulator of crowded colonies.* Growing RODSIM colonies to 4,000–9,000 cells reproduces OIAM's regime statistics (windows with >400 ancestors: 28% *vs.* 22% on OIAM and 10% in RODSIM v1; median displacement there 3.0 *vs.* 2.7 and 1.6 spacings). With all arms at the same learning rate, v3 does not improve zero-shot accuracy on OIAM (V1, V2 fail; dense windows lose 0.01) and does not harm the external bacterial datasets (V4); on dense OIAM windows the v3-trained centre beats the v3-trained direct matcher at 33 and 66 minutes (V3). Within v3 itself both models reach only 0.63 (0.29 above 400 ancestors), so covering the regime is not sufficient: the crowded regime is also where the task is hardest to learn from positions alone. *One extra snapshot.* The cheapest remedy is experimental: composing the assignment through one snapshot halfway through the window, $P(a \mid b) = P(m \mid b)\, P(a \mid m)$, raises the ensemble from 0.490 to 0.725 at 132 minutes and from 0.807 to 0.949 at 66 (A1). Label-free confidence ranks windows by accuracy (Spearman 0.93–0.97 between window-mean confidence and accuracy) but, as a policy for where to add the snapshot, recovers 79% of the gain with half the extra snap-

shots at 66 minutes and only 51% (no better than random) at 132 (A2 fails): at long gaps the network is overconfident everywhere (expected calibration error 0.27 *vs*. 0.01 at 33 minutes).

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Protocol	Data	Hypotheses (fixed before running)	Outcome (run once, reported as is)
Blind test 1	TrackRefiner, TrackMate; 11 seq., unseen	H1 RODSIM > best geometric, per source; H2 on most sequences; H3 RODSIM > OIAM-trained	H1 **fails** 0/2; H2 **fails** (2/11); H3 holds 2/2
Blind test 2	LYN yeast, *B. subtilis*; fixed before download	H1 SV > best geometric; H2 on most sequences; H3 SV $\geq$ both components	yeast: H1, H3 hold, H2 fails (3/6); *B. subtilis*: all **fail** (9/44)
Protocol 3	four blind-test sources, already scored; two folds	A1 adapted SV > best geometric; A2 > unadapted SV; A3 adapted > RODSIM	A1 3/4; A2 3/4; A3 4/4
Protocol 4	FAST, recA, rpsM (not blind)	B1 Hungarian self-labelling > RODSIM; B2 > best geometric	B1 **fails** 0/3; B2 1/3
Protocol 5‡	FAST, recA, rpsM (not blind)	C1 bootstrapping > RODSIM; C2 > best geometric	C1 3/3; C2 3/3
Protocol 6	as protocol 3	E1 bootstrapping > Hungarian self-labelling; E2 its SV > best geometric; E3 > SV with Hungarian	E1 **fails** 0/4; E2 3/4; E3 3/4
Blind test 4	trpL, metA, pheA; 28 colonies, fixed before download	F1 RODSIM > best of six baselines; F2 on most colonies; F3 bootstrapping > RODSIM; F4 > Hungarian	all hold, 3/3 subsets, every gap
Teacher reliability	ten self-labelling sources, 114 units	(a) label-free teacher statistics track one-frame link accuracy; (b) primary statistic separates sources where bootstrapping wins	(a) holds (ρ=0.84); (b) **fails** (2/10 wrong)
Matched direct matcher	OIAM M3/M4; 80k steps, 3 seeds per arm	D1 LineageRAFT > direct matcher (Darcy) every gap; D2 dense; D3 radial init.	D1, D2 hold (CIs >0); D3 **fails** at 132 min (tie)
Trackastra without rot. aug.	OIAM M3/M4; batch 8 (amended after a batch-16 run)	R1 no-rot. fine-tuning helps Trackastra at $\geq$2 gaps; R2 LineageRAFT > better Trackastra at 33/66; R3 decoded ensemble > it at all gaps	R1 **fails** (1/3); R2, R3 hold
Full-model LOMO	five OIAM folds, 80k steps, one seed	L1 > no-rot. 30k at 33/66 on $\geq$4/5 folds; L2 > regressor at every gap on $\geq$4/5; L3 mean $\geq$0.90 at 33	all hold (5/5, 5/5; 0.954)
Centre controls	OIAM M3/M4; 80k steps, 3 seeds per arm	K1 LineageRAFT > direct matcher with 96 candidates; K2 > centre loss without moving; K3 T=4 > T=1	all hold (CIs >0 at 33/66 min)
Orientation	OIAM M3/M4 in 8 image frames; estimator chosen on M0–M2	H1 no correction worse; H2 primary rule recovers $\geq$75% of the no-rotation gain; H3 per-window < per-sequence; H4 confidence > averaging	H1 holds; H2 holds at 33/66 (nothing to recover at 132); H3 mixed; H4 **fails**
Decoder on baselines	OIAM M3/M4, decoder windows	G1 decoded LineageRAFT > decoded Trackastra at every gap (CI >0 at 33/66); G2 > decoded GBDT; G3 decoded ensemble > both at 132	all hold; *vs*. raw-probability Trackastra +0.010 at 132 (CI includes 0)
Time-block bootstrap	OIAM M3/M4, 7 main comparisons	B1 every window-significant margin stays significant with blocks of 4 windows	**fails**: 14/19; 132-min single-model margins and 66-min decoder-on-decoder lost
Optical flow	OIAM; RAFT setting chosen on M0–M2 (8 settings)	R1 LineageRAFT > flow at every gap; R2 flow > nearest at 33 min	both hold (flow 0.636/0.367/0.244)
Trackastra zero-shot	FAST (images + corrected masks), 7 colonies	Z1 RODSIM-trained LineageRAFT > Trackastra; Z2 on $\geq$5/7 colonies	both hold (0.709 *vs*. 0.626; 7/7)
RODSIM direct matcher	zero-shot table, 4 bacterial + 2 human datasets; 3 seeds (rerun at half lr after divergence)	S1 LineageRAFT > direct matcher on all 4 bacterial datasets; S2 direct matcher > best baseline on $\geq$3/4	S1 **fails** (2/4: OIAM, FAST); S2 holds (4/4)
Matched Trackastra	OIAM M3/M4; LineageRAFT on M0/M1, 20k steps, 3 seeds; Trackastra over 5 gaps	M1/M2 matched LineageRAFT > per-gap / multi-gap Trackastra at 33/66; M3 $\geq$ selected Trackastra at 132	all **fail** (0.715/0.554/0.394); multi-gap Trackastra replaces the 132-min entry (0.488)
Direct matcher, van Vliet	28 colonies of blind test 4; bootstrapped self-labelling, 6k steps	V1 LineageRAFT > direct matcher over colonies (CI > 0); V2 in every subset	V1 fails (+0.003 [−0.001, 0.007]); V2 holds
Self-labelled direct matcher	OIAM M3/M4; pseudo-lineage of M0–M2, 80k steps, 3 seeds	P1 self-labelled LineageRAFT > self-labelled direct matcher at 33/66; P2 self-labelling costs the direct matcher <0.05	both hold (+0.088/+0.059; cost 0.01)
Global partitioners	OIAM M3/M4, 66/132 min	no threshold; correlation and spectral clustering *vs*. Potts	predicted affinities: no gain (0.517 *vs*. 0.517); true: 0.780 *vs*. oracle 0.784
Self-labelled genealogy	OIAM M3/M4	no threshold; ensemble decoder *vs*. annotated genealogy	0.976/0.861/0.520 *vs*. 0.977/0.863/0.517
Annotation-free pipeline	OIAM M3/M4	no threshold; full model on pseudo-lineage, 3 seeds	0.963/0.822/0.487; no-consistency variant post hoc
Fine-tuned Trackastra	OIAM; trained on M0/M1, selected on M2	no threshold; one model per gap, decoding fixed on M2	0.892/0.716/0.451 (LineageRAFT 0.957/0.780/0.473)
Lineage audit	recA, rpsM *vs*. curated FAST	no threshold; descriptive label statistics	flagged targets 5.2%/1.7% *vs*. 0.2% (correction post hoc)
Second segmenter	Omnipose on OIAM M3/M4, notebook settings	no threshold; end-to-end accuracy	0.74/0.50/0.29 (*vs*. 0.92/0.73/0.44 Cellpose-SAM); dense loss $\leq$0.08
Temperature scaling	OIAM; T fitted on M0–M2	no threshold; decoded once on M3/M4	single seeds +0.010–0.014 at 66 min; ensemble unchanged
Structured decoder	OIAM M4 (M3 for development); FAST, recA	vs Potts at 66/132: H1 lower size error and singleton rate; H2 pair F1; H3 accuracy not	H1, H2 hold; H3 holds at 66, **fails** at 132; H4 **fails** (accuracy 0/3)

	All windows: mean [block-bootstrap 95% CI] (M3 / M4)			Dense ($N_a > 200$): mean [CI]		
Method	33	66	132	33	66	132
Nearest ancestor	.497 [.312, .678] (.48/.51)	.361 [.196, .513] (.34/.38)	.263 [.143, .368] (.26/.27)	.157 [.127, .179]	.083 [.071, .091]	.058 [.054, .060]
Radial expansion	.547 [.392, .703] (.57/.52)	.458 [.318, .606] (.44/.48)	.342 [.230, .467] (.34/.35)	.249 [.137, .354]	.154 [.059, .225]	.056 [.023, .090]
Darcy flow prior (ours)	.627 [.477, .802] (.62/.64)	.466 [.327, .614] (.44/.49)	.368 [.239, .494] (.35/.39)	.305 [.172, .450]	.161 [.087, .240]	.088 [.064, .122]
Optical flow (RAFT [8])	.636 [.500, .789] (.61/.66)	.367 [.230, .502] (.35/.39)	.244 [.129, .349] (.24/.25)	.347 [.256, .460]	.094 [.059, .123]	.041 [.034, .052]
Unbalanced OT [7]	.555 [.400, .712] (.58/.53)	.473 [.328, .628] (.45/.50)	.353 [.234, .480] (.34/.36)	.257 [.144, .367]	.159 [.060, .232]	.056 [.024, .088]
Non-rigid CPD [6]	.597 [.445, .757] (.60/.60)	.466 [.328, .616] (.44/.50)	.359 [.243, .490] (.34/.38)	.294 [.154, .418]	.162 [.063, .235]	.060 [.025, .096]
Family-centre regression (GBDT)	.731 [.612, .880] (.74/.72)	.610 [.472, .772] (.60/.62)	.446 [.328, .561] (.43/.46)	.449 [.372, .567]	.297 [.252, .360]	.201 [.175, .231]
Trackastra [4], argmax	.681 [.553, .843] (.69/.67)	.176 [.096, .245] (.14/.22)	.162 [.084, .213] (.15/.18)	.395 [.295, .523]	.073 [.070, .081]	.057 [.054, .059]
Trackastra, fine-tuned†	.892 [.855, .954] (.90/.89)	.716 [.638, .828] (.72/.71)	.488 [.396, .609] (.49/.49)	.788 [.753, .865]	.507 [.445, .615]	.230 [.176, .314]
LineageRAFT, rot. aug.	.741 [.612, .899] (.74/.75)	.604 [.474, .759] (.59/.62)	.451 [.343, .573] (.45/.45)	.459 [.330, .604]	.294 [.205, .390]	.178 [.137, .223]
LineageRAFT, no rot.	.853 [.792, .928] (.87/.84)	.653 [.563, .761] (.65/.65)	.447 [.358, .539] (.45/.44)	.717 [.675, .775]	.443 [.400, .495]	.247 [.228, .276]
Direct matcher (LineageRAFT, fixed centre)	.856 [.803, .944] (.87/.85)	.684 [.589, .815] (.68/.69)	.456 [.355, .567] (.45/.47)	.706 [.656, .819]	.429 [.364, .536]	.207 [.180, .252]
LineageRAFT, full	.957 [.940, .986] (.96/.95)	.780 [.710, .873] (.78/.78)	.473 [.382, .563] (.46/.48)	.907 [.890, .952]	.598 [.547, .674]	.267 [.248, .296]
LineageRAFT, ensemble + decoder	.977 [.967, .996] (.98/.97)	.863 [.808, .950] (.86/.87)	.517 [.424, .616] (.51/.52)	.946 [.935, .979]	.703 [.651, .785]	.300 [.280, .332]
LineageRAFT, ensemble + decoder, no annotation	.963 [.947, .990] (.97/.96)	.822 [.753, .917] (.81/.84)	.487 [.385, .591] (.48/.50)	.919 [.902, .960]	.634 [.579, .719]	.260 [.234, .295]
Oracle family partition	.984 [.977, .999] (.99/.98)	.929 [.895, .994] (.93/.93)	.784 [.670, .928] (.76/.81)	.963 [.952, .988]	.817 [.767, .899]	.465 [.442, .516]

Table S22. **Table 2 under one protocol.** Every row recomputed from its per-window results by the code that writes Tab. 2, on the same windows (39/37/33 windows at 33/66/132 minutes, 16/14/10 dense; at most 1 missing and 0 extra windows in any row). CIs: time-block bootstrap, blocks of four consecutive windows within each movie (5000 resamples), which respects the overlap of windows; in parentheses the means over M3 and M4 separately. Seeded rows: per-window means over three seeds. Windows, not movies, are the resampling unit because only two test movies exist; the per-movie means show the between-movie spread.

	All windows			Dense (N_a>200)		
Intervention	33	66	132	33	66	132
Network (ensemble argmax)	.964	.807	.490	.923	.634	.284
+ true family centres	.989	.919	.605	.975	.845	.498
+ true partition	.984	.929	.786	.963	.817	.468
+ both	.998	.993	.965	.994	.982	.903
True ancestor among the 24 candidates	.998	.992	.984	.996	.980	.946

Table S23. **Which oracle fixes long gaps?** OIAM M3/M4, three-seed ensemble of the full model, window-mean accuracy. True centres: the final family centre of every ancestor is replaced by the centroid of its true family and the network re-scores its candidates (features unchanged). True partition: the descendant partition is given to the decoder. Protocol fixed before the run.

	Accuracy		Dense acc.		Pair F1		Size err.
Decoder	66	132	66	132	66	132	132
Argmax	.808	.501	.614	.283	.663	.323	0.96
Potts (main paper)	.868	.523	.682	.294	.821	.391	0.91
Growth prior only	.822	.493	.621	.274	.709	.305	0.59
Pairwise likelihood only	.880	.528	.691	.295	.870	.421	0.82
Structured	.883	.525	.697	.282	.880	.422	0.70
Structured + centre update	.883	.513	.702	.285	.885	.425	0.71
True family sizes	.892	.526	.718	.312	.880	.409	0.07
True partition	.926	.813	.782	.448	1.000	1.000	0.20

Table S24. **A family-size prior repairs grouping, not long-gap ancestry** (OIAM M4, the confirmatory movie; same ensemble scores and genealogy affinities for every decoder; protocol and hyperparameters fixed before M4 was decoded). Size err.: mean $|n_{\text{pred}} - n_{\text{true}}|$ per ancestor; truth has std 0.81 of family sizes at 132 minutes.

	Median		Margin by ancestors per window			
Test windows, training	N_a	disp.	≤25	26–100	101–400	>400
OIAM, trained on M0–M2	108	2.0	+0.011 (27)	+0.025 (27)	+0.078 (30)	+0.190 (25)
OIAM, trained on RODSIM	108	2.0	+0.015 (27)	+0.026 (27)	+0.018 (30)	+0.016 (25)
FAST/recA/rpsM, RODSIM	4	1.9	-0.005 (2173)	-0.002 (361)	+0.011 (12)	–
T98G/LFCT, RODSIM	9	0.2	+0.012 (1141)	+0.023 (88)	–	–

Table S25. **Where the moving centre helps** (post hoc). Per-window accuracy of LineageRAFT minus the direct matcher trained identically (three seeds each; in brackets the number of windows), all gaps pooled. Disp.: median displacement of targets from their true ancestor after removing the mean shift, in anchor spacings. The external bacterial windows are small; on crowded OIAM windows the centre learned from RODSIM adds little.

Comparison	Windows	33 min	66 min	132 min
Coupled − LineageRAFT (OIAM, 3+3 seeds)	all	-0.004 [-0.008, +0.001]	-0.009 [-0.027, +0.006]	+0.000 [-0.010, +0.020]
	dense	+0.003 [+0.001, +0.008]	+0.021 [+0.015, +0.031]	+0.012 [+0.005, +0.020]
RodSim v3 − v1, zero-shot (OIAM)	all	-0.001 [-0.009, +0.008]	+0.000 [-0.006, +0.008]	-0.005 [-0.008, -0.001]
	dense	-0.001 [-0.011, +0.003]	-0.008 [-0.015, -0.006]	-0.010 [-0.013, -0.007]
RodSim v3: LineageRAFT − direct matcher	dense	+0.024 [+0.013, +0.039]	+0.007 [+0.007, +0.013]	+0.003 [-0.002, +0.010]
Midpoint snapshot, ensemble	all	–	0.807→0.949 (+0.142 [+0.109, +0.177])	0.490→0.725 (+0.235 [+0.209, +0.260])
	dense	–	0.634→0.890 (+0.257 [+0.221, +0.289])	0.284→0.500 (+0.217 [+0.177, +0.251])

Table S26. **Three attempts at long gaps**, each with hypotheses fixed before running (Tab. S21). Window-mean argmax accuracy differences with block-bootstrap 95% CIs (window bootstrap for the snapshot rows); dense: >200 ancestors. Coupled: E/M competition of ancestors for targets inside each refinement round, same recipe as LineageRAFT otherwise. RodSim v3: colonies of 4,000–9,000 instead of 900–2,200 cells, all arms at learning rate 2.5×10^{-4}. Midpoint snapshot: assignment composed through one extra snapshot halfway through the window.